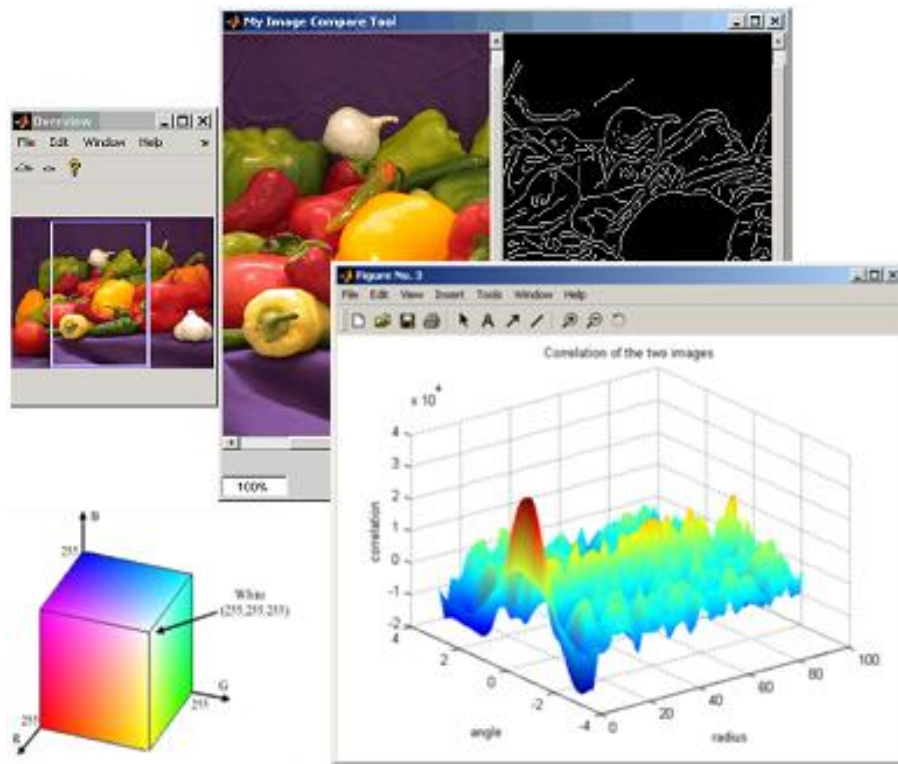


Digital Image Processing

LECTURE 14: MORPHOLOGICAL OPERATION (PART 2)



Dr. Ajay Kumar Mahato
Assistant Professor
ECE Department
GLA University Mathura

Morphological Operations (Recap)

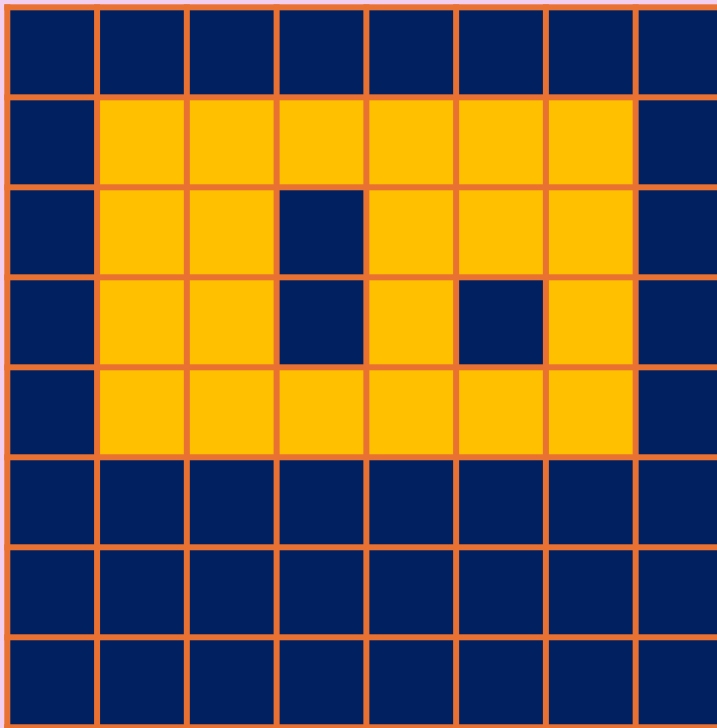
Morphological Operations

- ❑ Image Morphology is the study of shapes of the objects present in the image and extraction of image features. Image features are necessary for object recognition.
- ❑ We use the same word here in the context of mathematical morphology as a tool for **extracting image components** that are useful in the **representation and description of region shape**, such as **boundaries, skeletons** and **convex hull**.

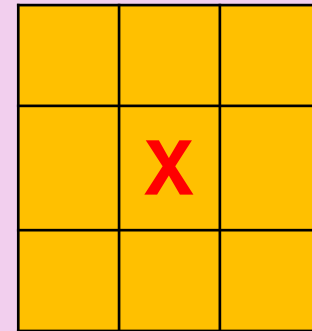
Dilation Operations

Perform $X \oplus B$

X

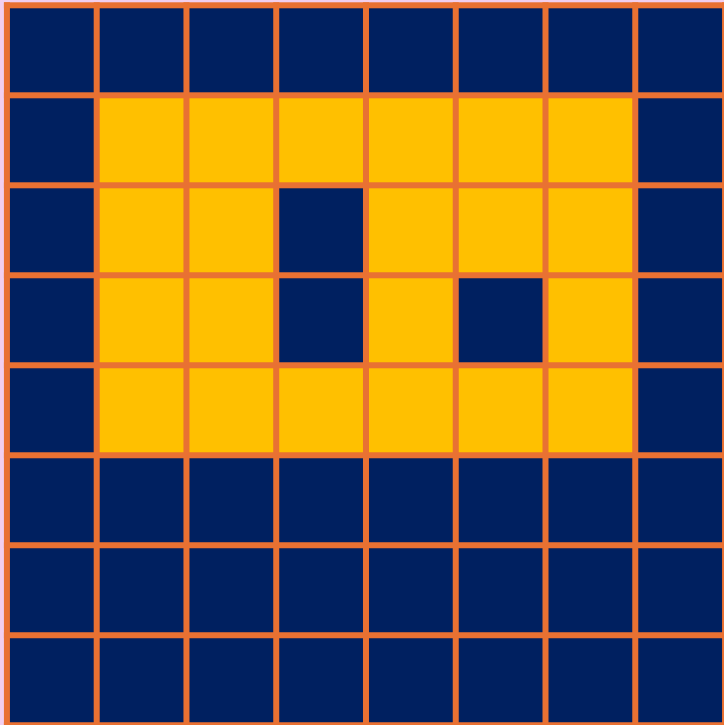


B

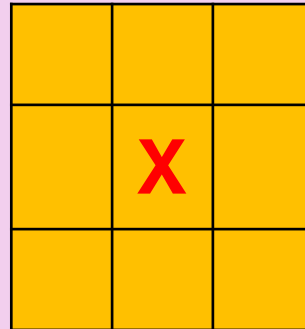


Dilation Operations

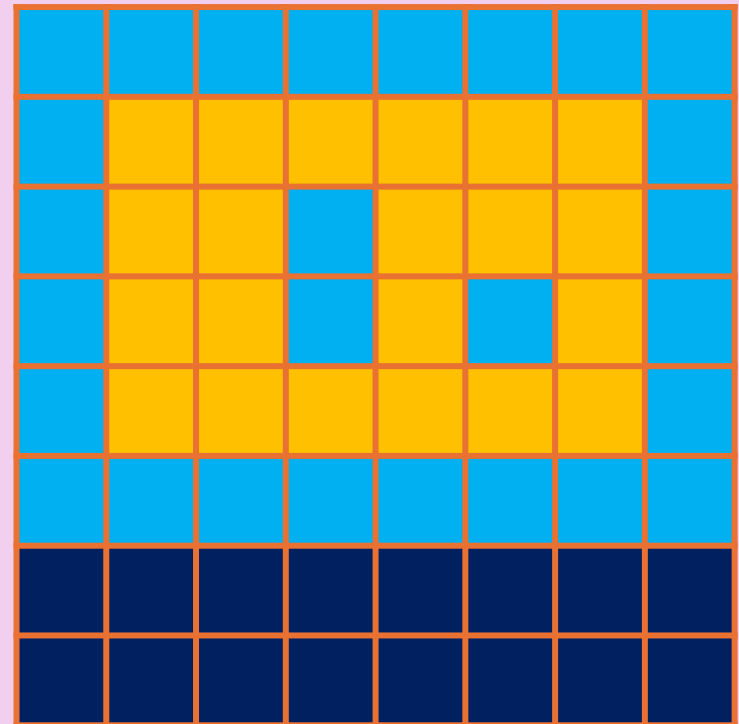
X



B



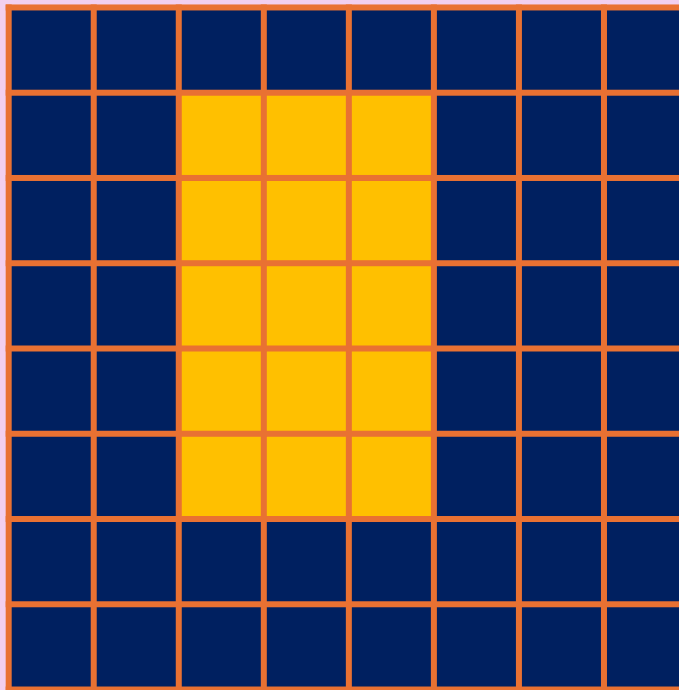
$X \oplus B$



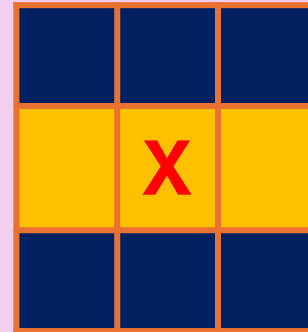
Erosion Operation

Perform $X \ominus B$

X

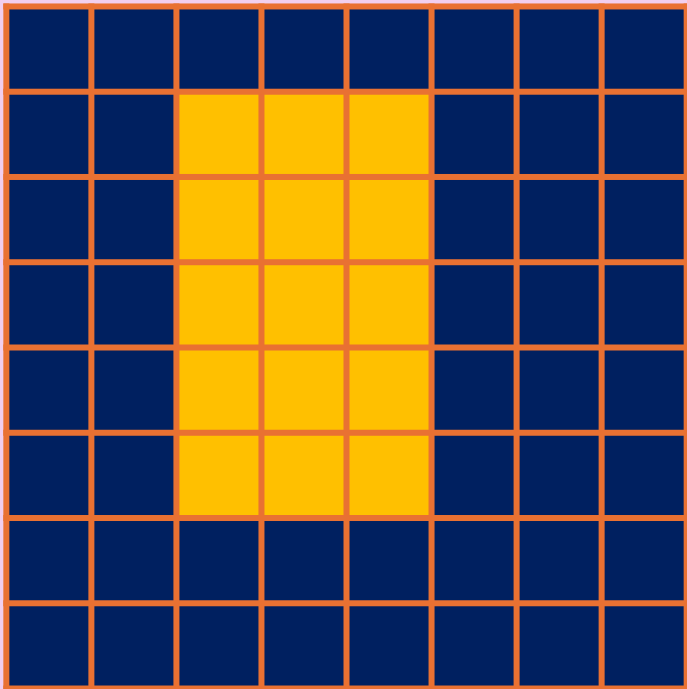


B

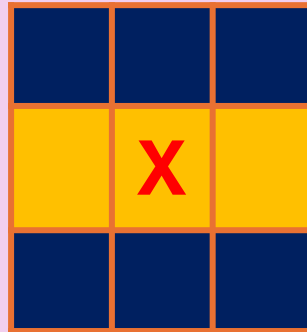


Erosion Operation

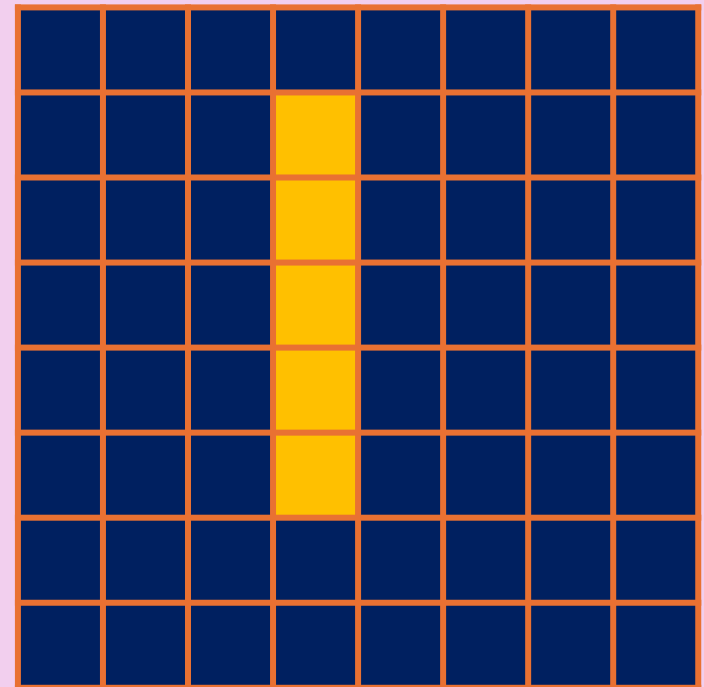
X



B



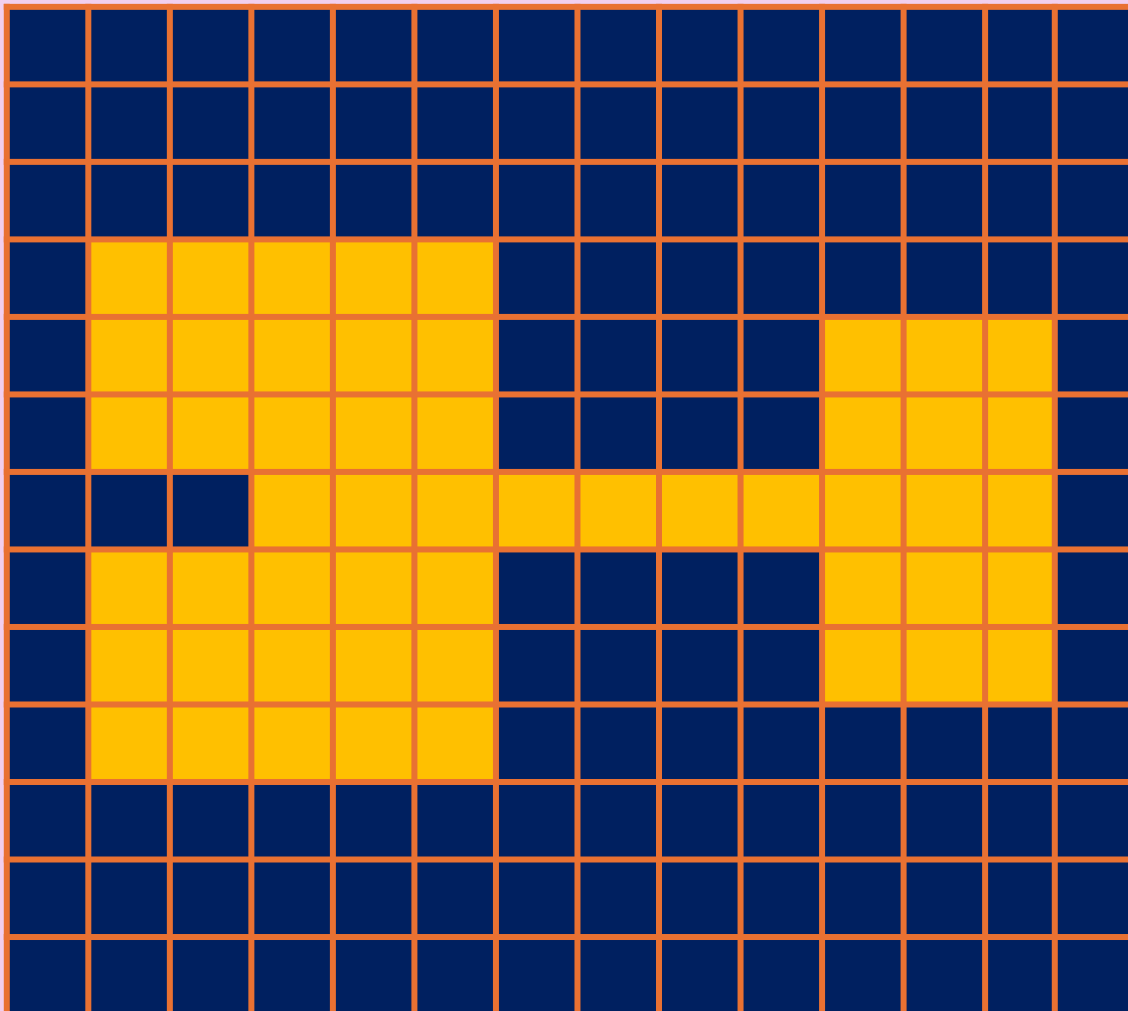
$X \ominus B$



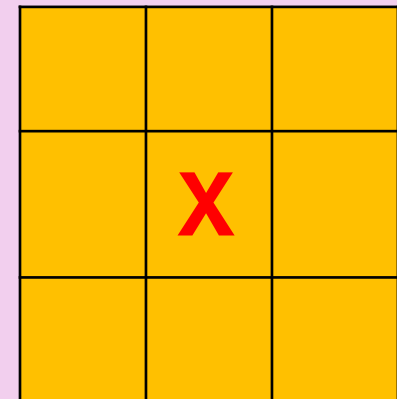
Opening Operation

□ Opening Operation: $A \circ B = (A \ominus B) \oplus B$

A



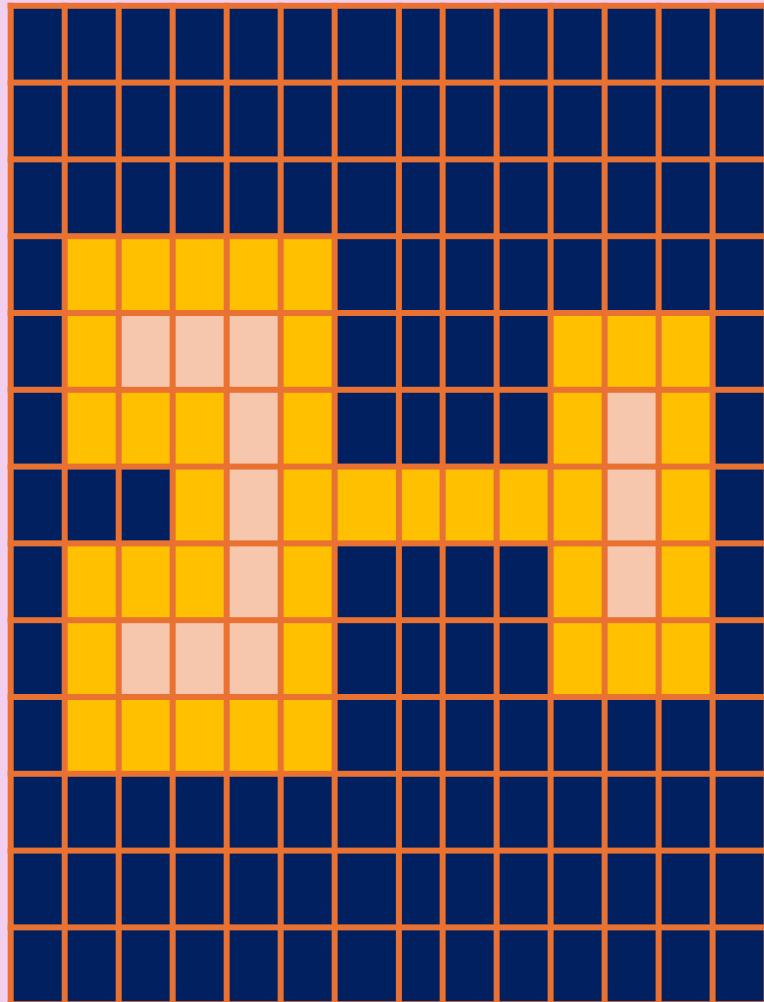
B



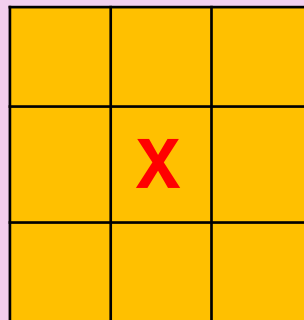
Opening Operation

$$A \circ B = (A \ominus B) \oplus B$$

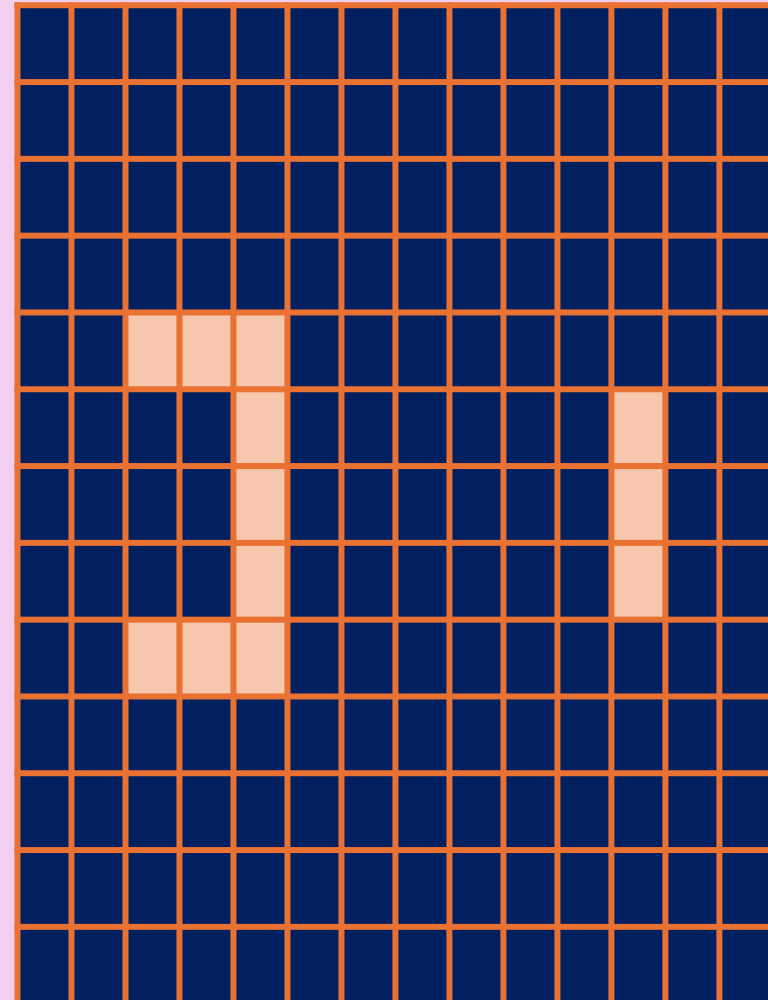
A



B



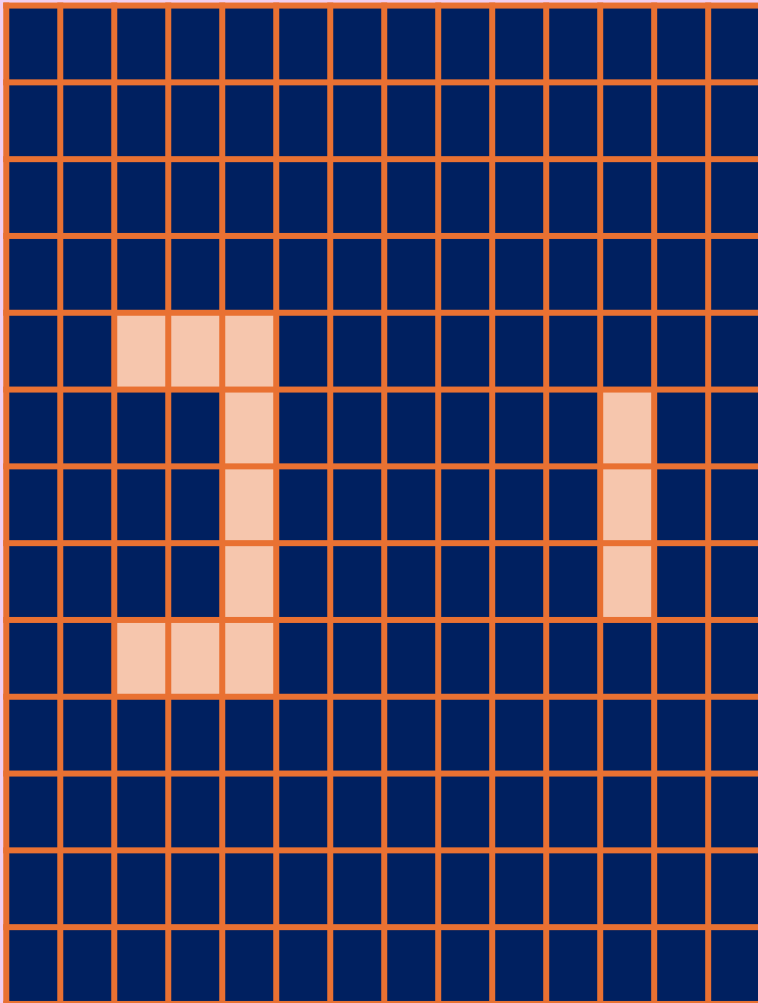
(A $\ominus$ B)



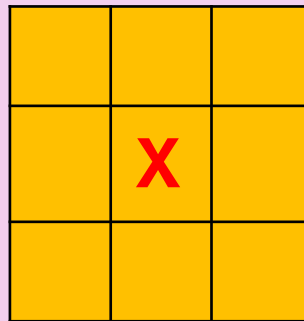
Opening Operation

$$A \circ B = (A \ominus B) \oplus B \quad \text{Opening}$$

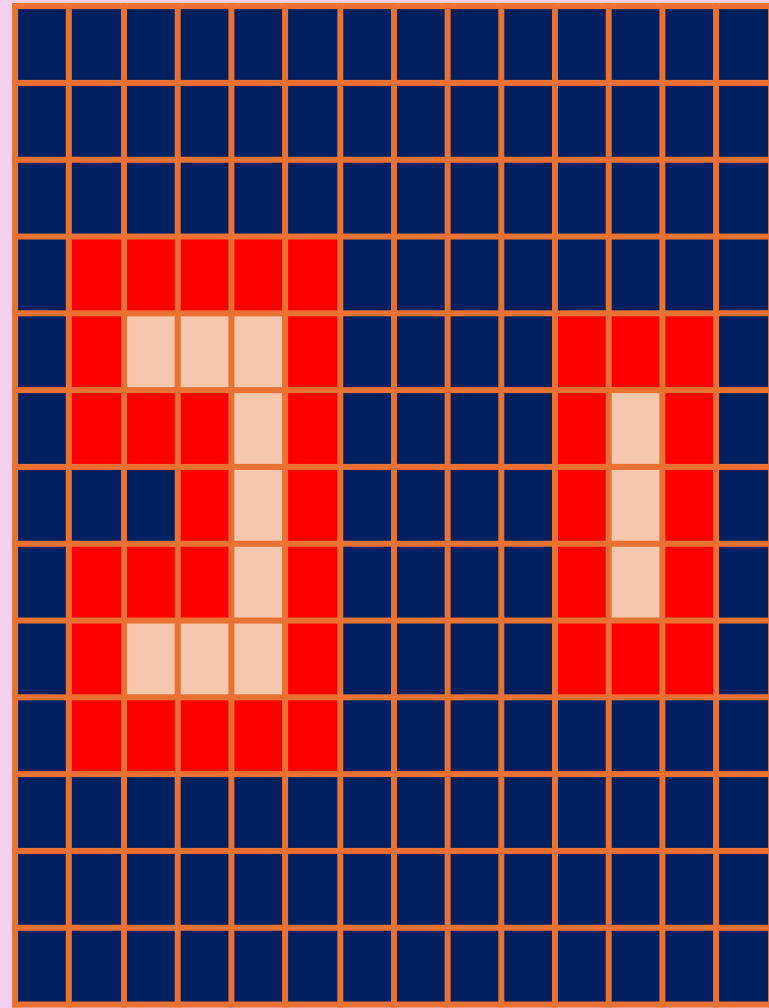
$$(A \ominus B)$$



B



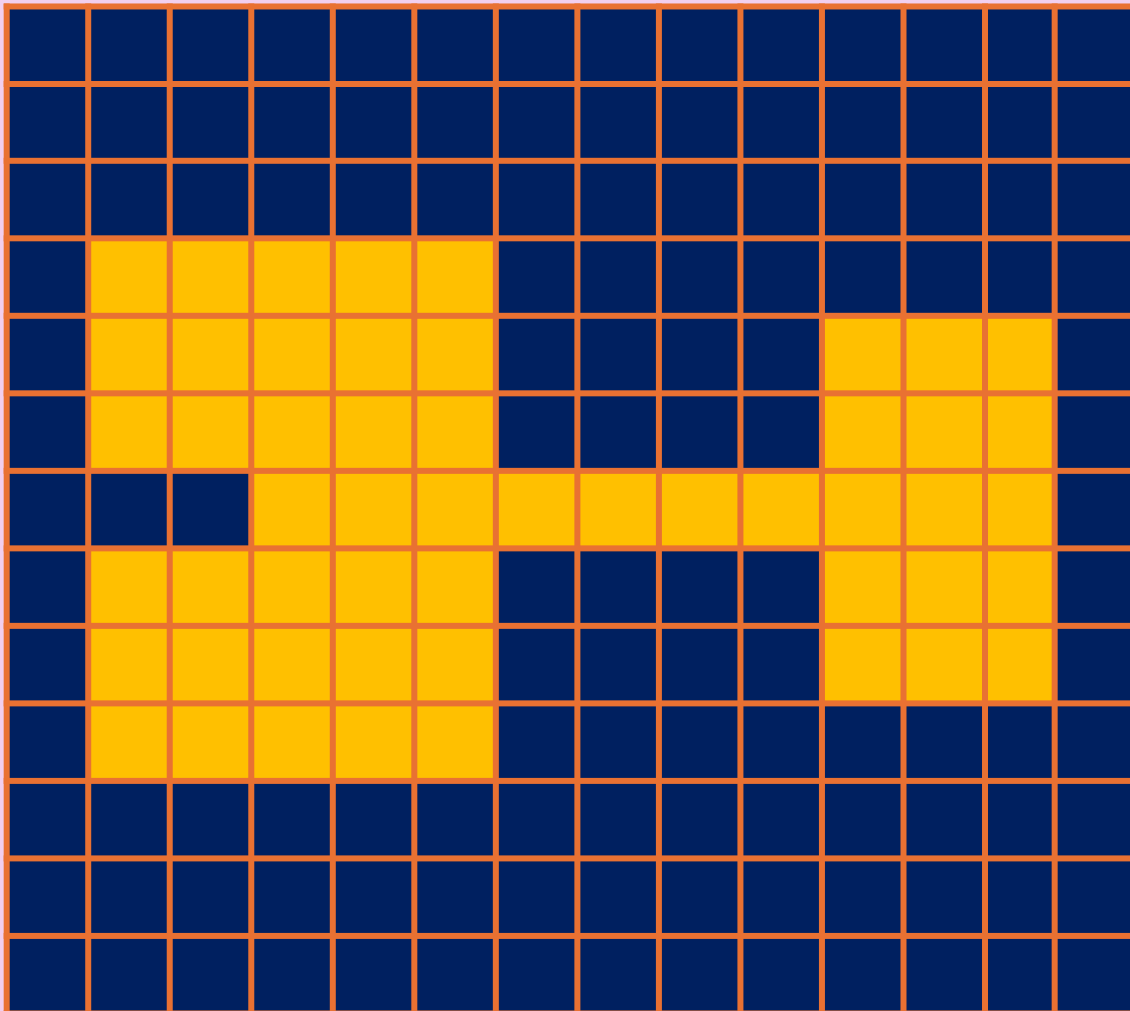
$$A \circ B = (A \ominus B) \oplus B$$



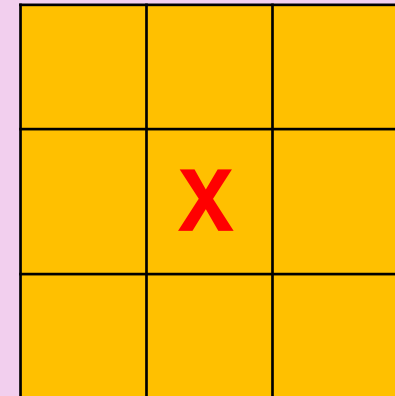
Closing Operation

□ Closing Operation: $A \cdot B = (A \oplus B) \ominus B$

A



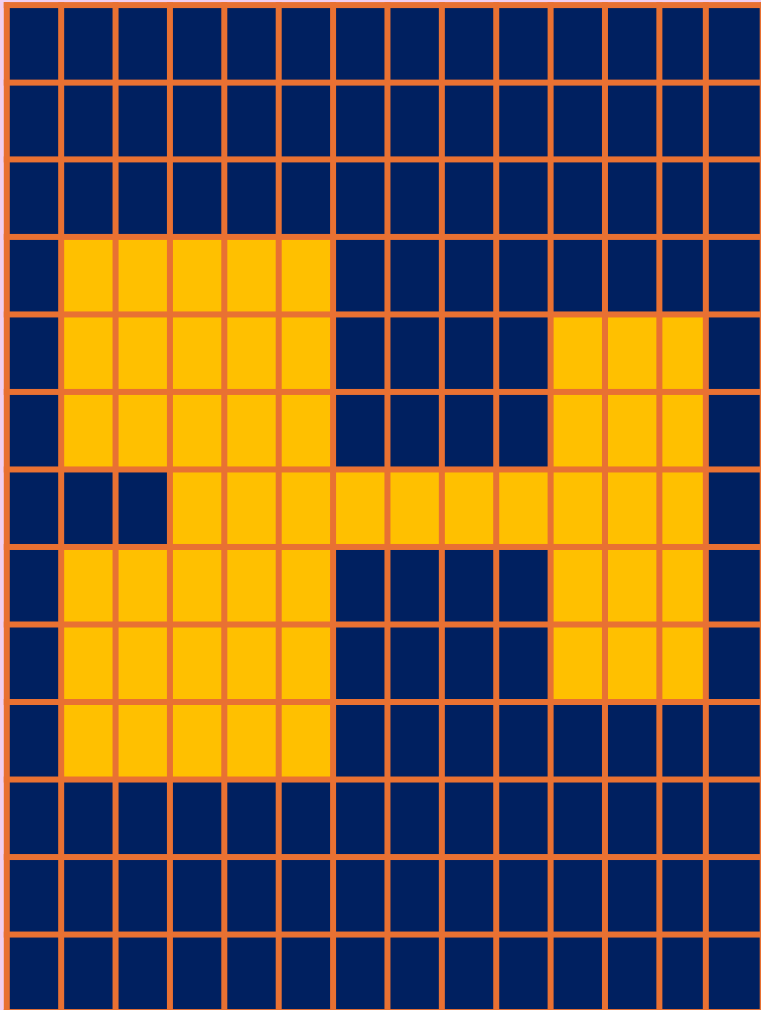
B



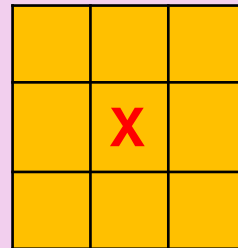
Closing Operation

$$A \cdot B = (A \oplus B) \ominus B$$

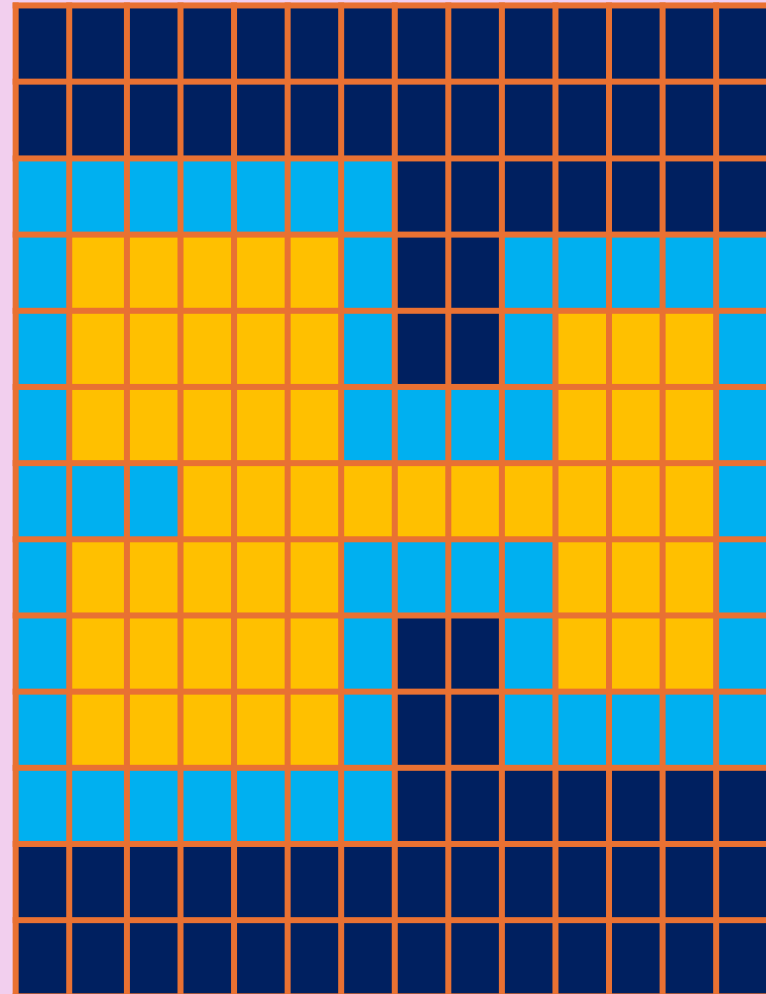
A



B



$(A \oplus B)$

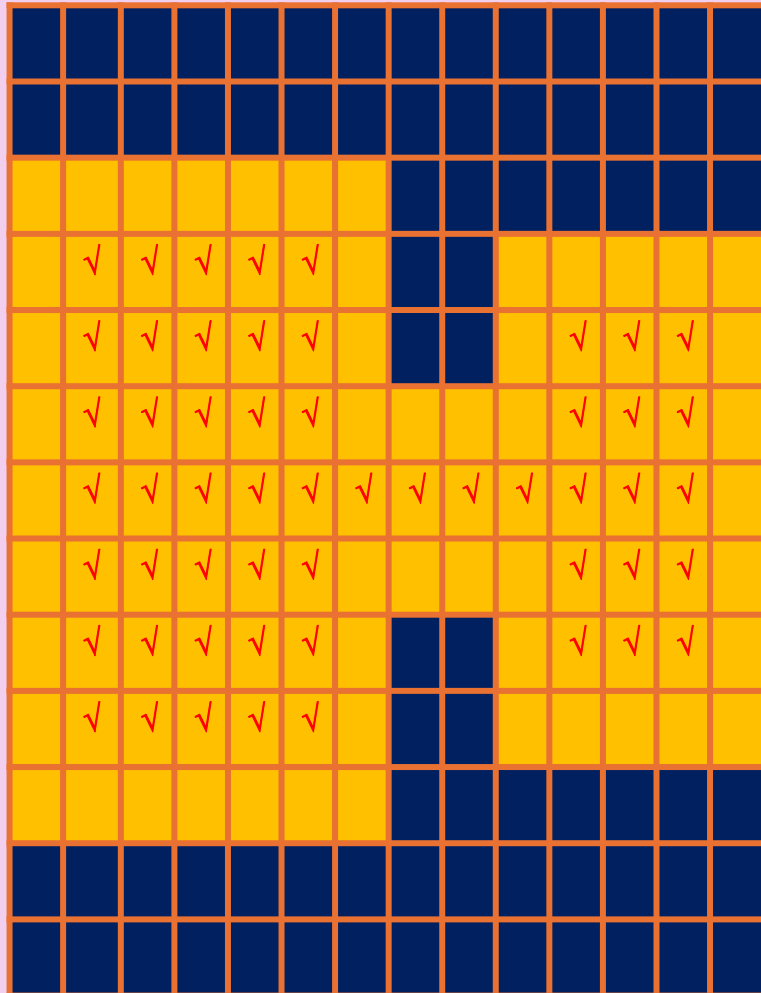


Closing Operation

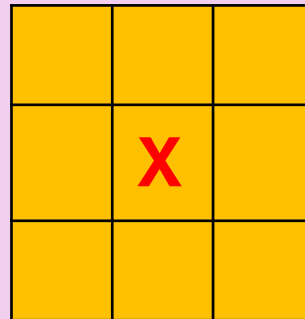
$$A \cdot B = (A \oplus B) \ominus B$$

Closing

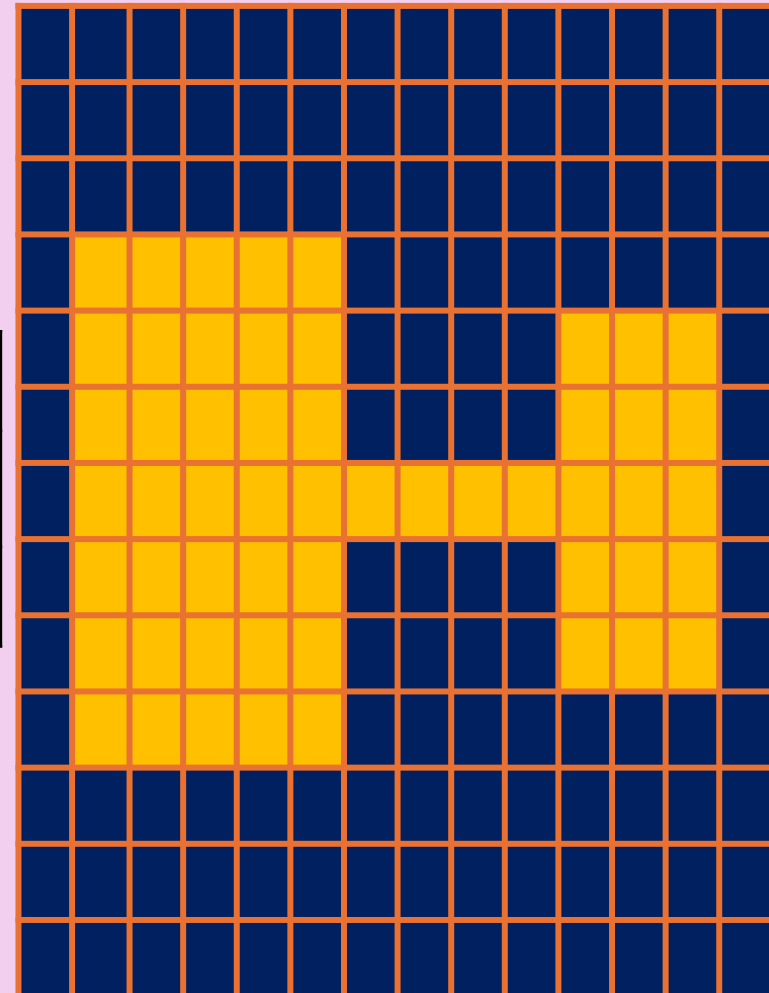
$$(A \oplus B)$$



B



$$A \cdot B = (A \oplus B) \ominus B$$



Hit or Miss Transform

Hit or Miss Transform

- The morphological hit-or-miss transform (HMT) is a basic tool for **Shape Detection**.
- It is useful to be able to match specified configurations of pixels in an image, such as isolated foreground pixels or pixels that are endpoints of line segment.
- It is a morphological operator **for finding local patterns of pixels**.
- The hit-or-miss transform is a general **binary morphological operation** that can be used to **look for patterns of foreground and background pixels in an image**

□ Concepts:

- Hit object
- Miss background

Hit or Miss Transform

❑ Mathematical expression for Hit or Miss Transform is given as:

$$I \circledast S = (I \ominus S) \cap (I^c \ominus (W - S))$$

❑ It can be written as

$$I \circledast S = (I \ominus S_1) \cap (I^c \ominus S_2)$$

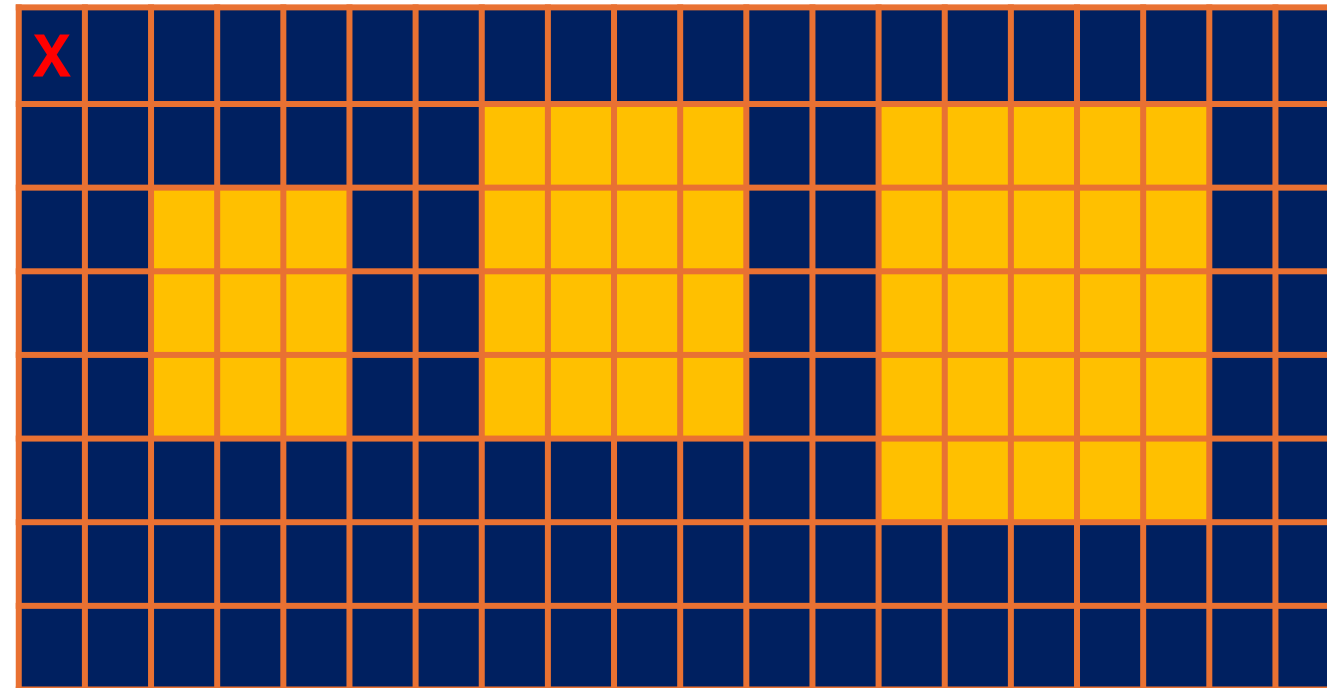
Where,

- S_1 is the set formed from elements of S associated with an object (S in this case)
- S_2 is the set of elements of S associated with the corresponding background ($W - S$)

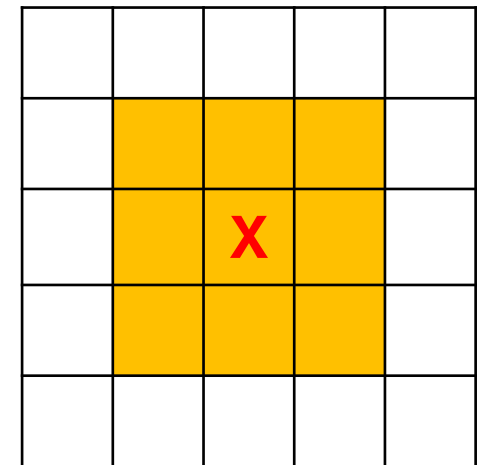
❑ **HIT Condition:** The set contains all the points at which, S_1 found a match (HIT) in I and S_2 found a match in I^c

Hit or Miss Transform

Example 1:



Input Image

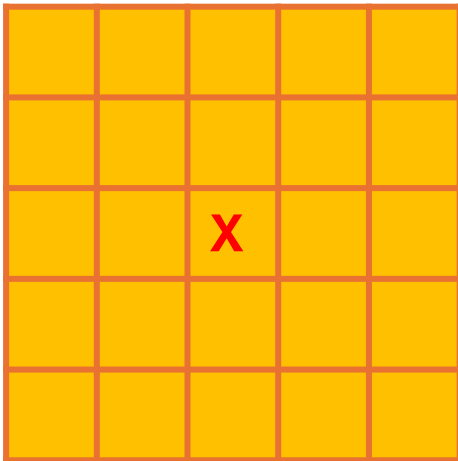


Structuring Element

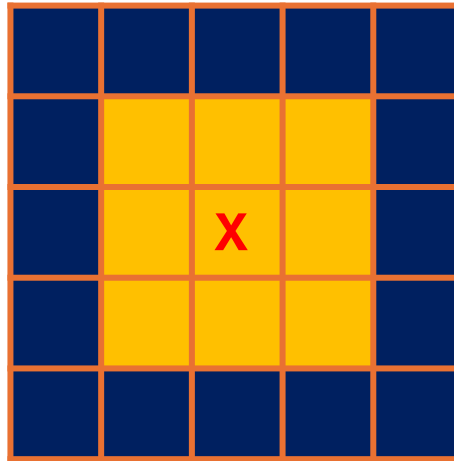
Hit or Miss Transform

Step 1:

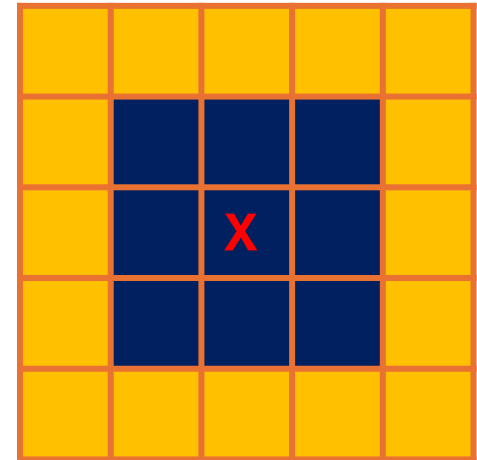
W



S



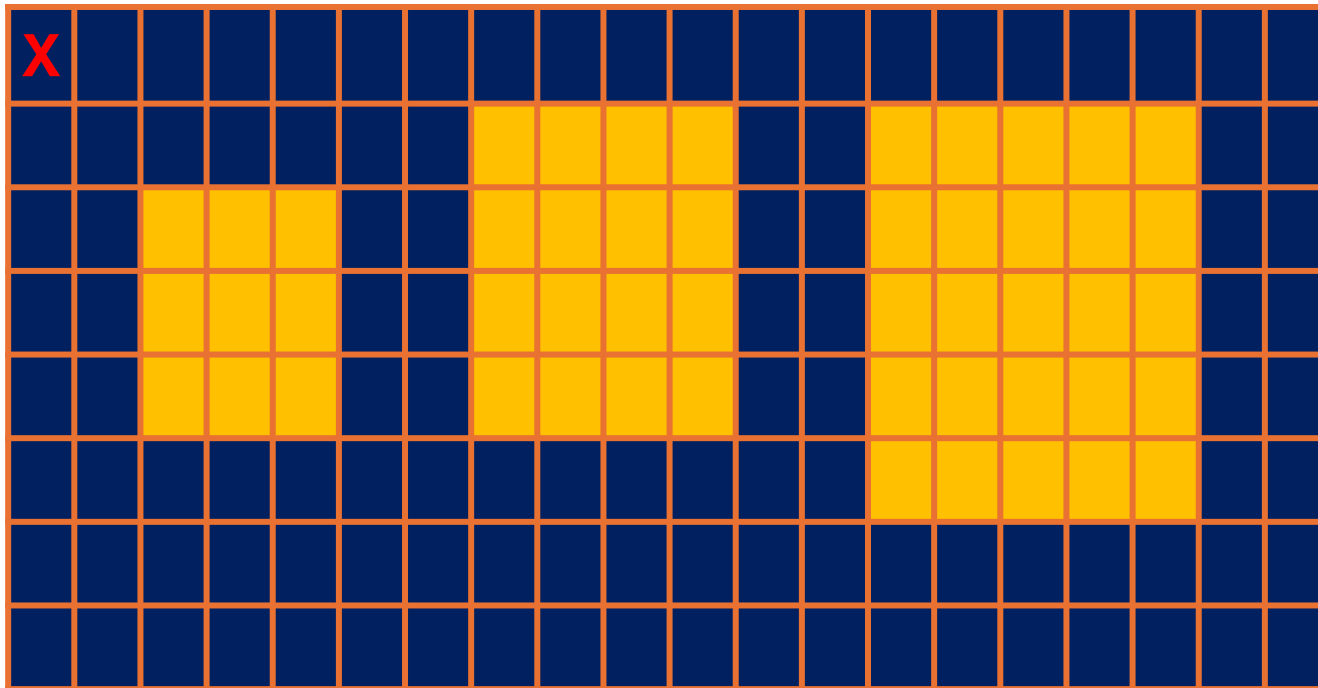
W- S



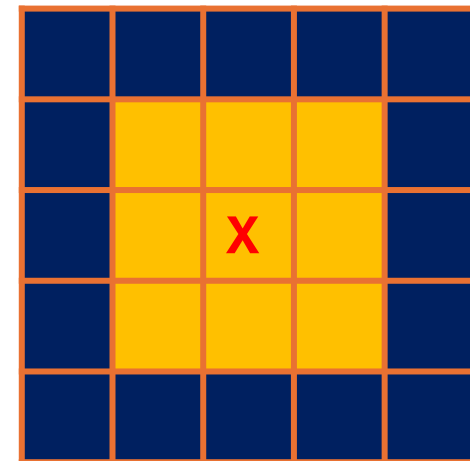
Hit or Miss Transform

Step 2: Perform the $(I \ominus S)$

I

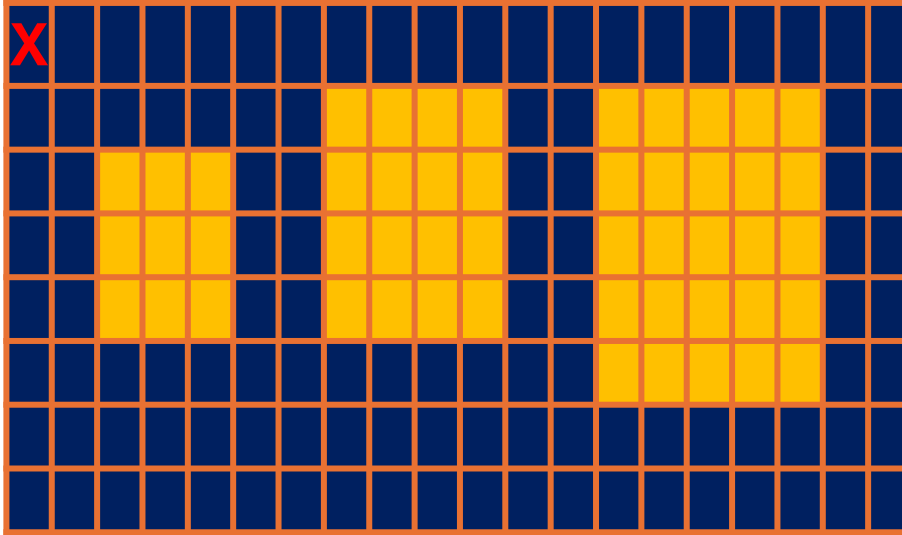


S

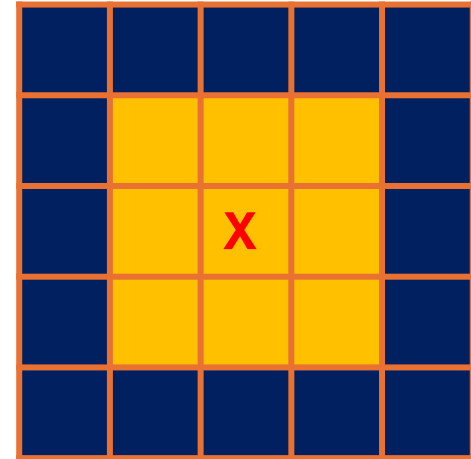


Hit or Miss Transform

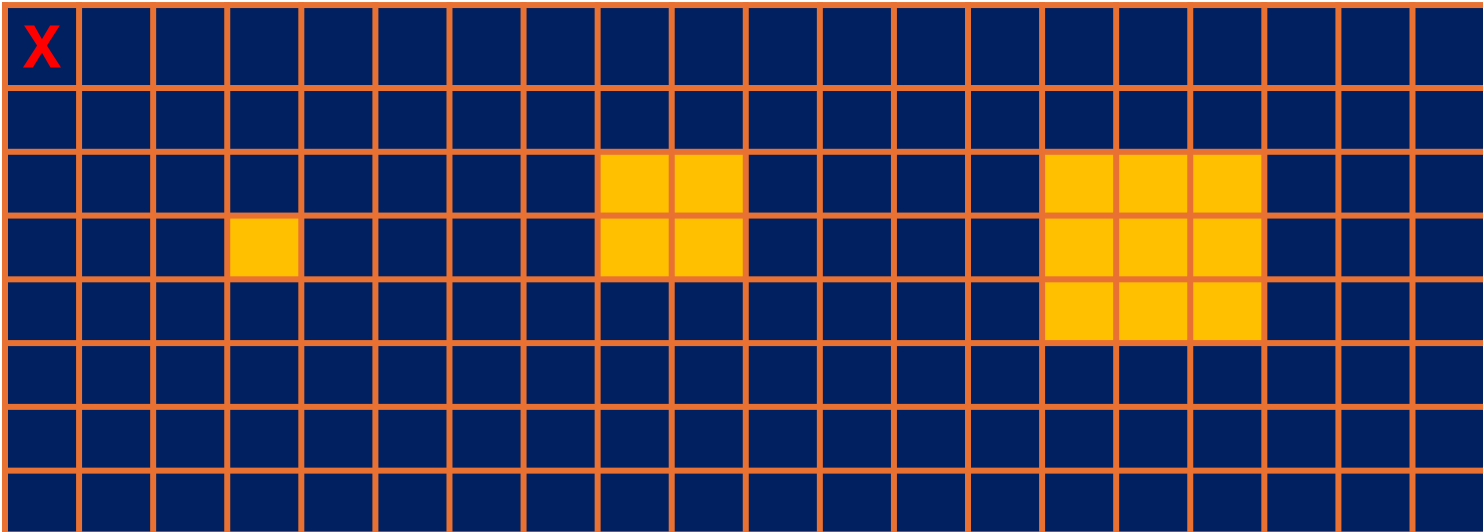
I



S



$I \ominus S$

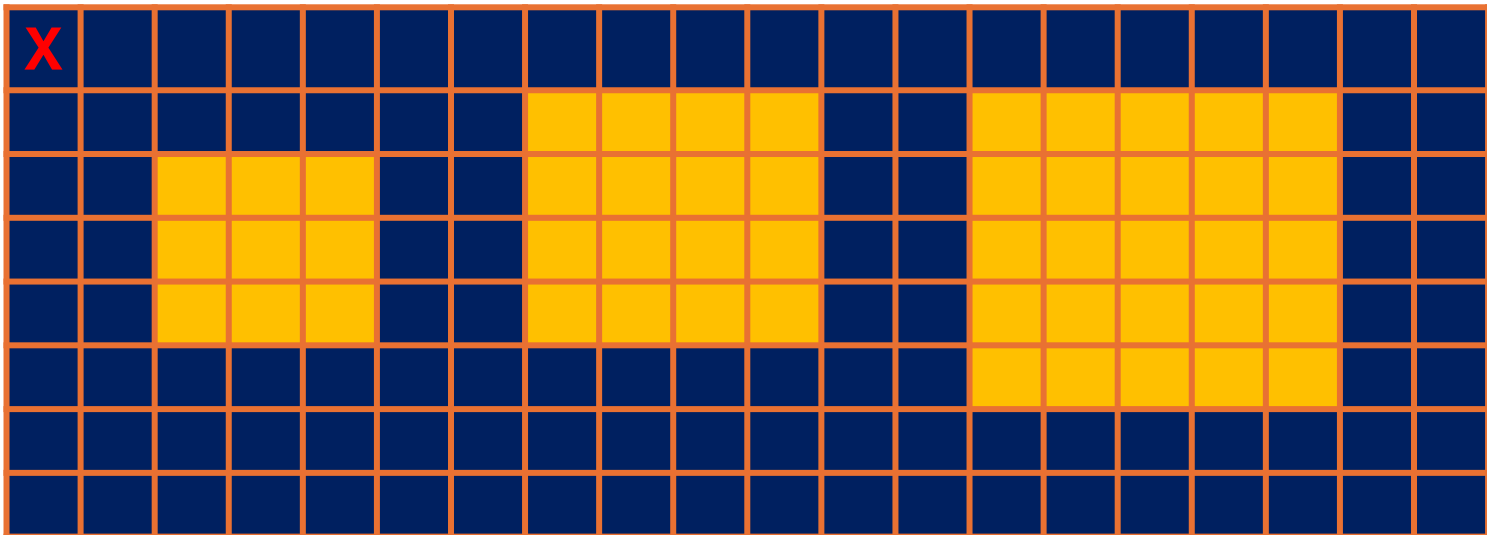


Hit or Miss Transform

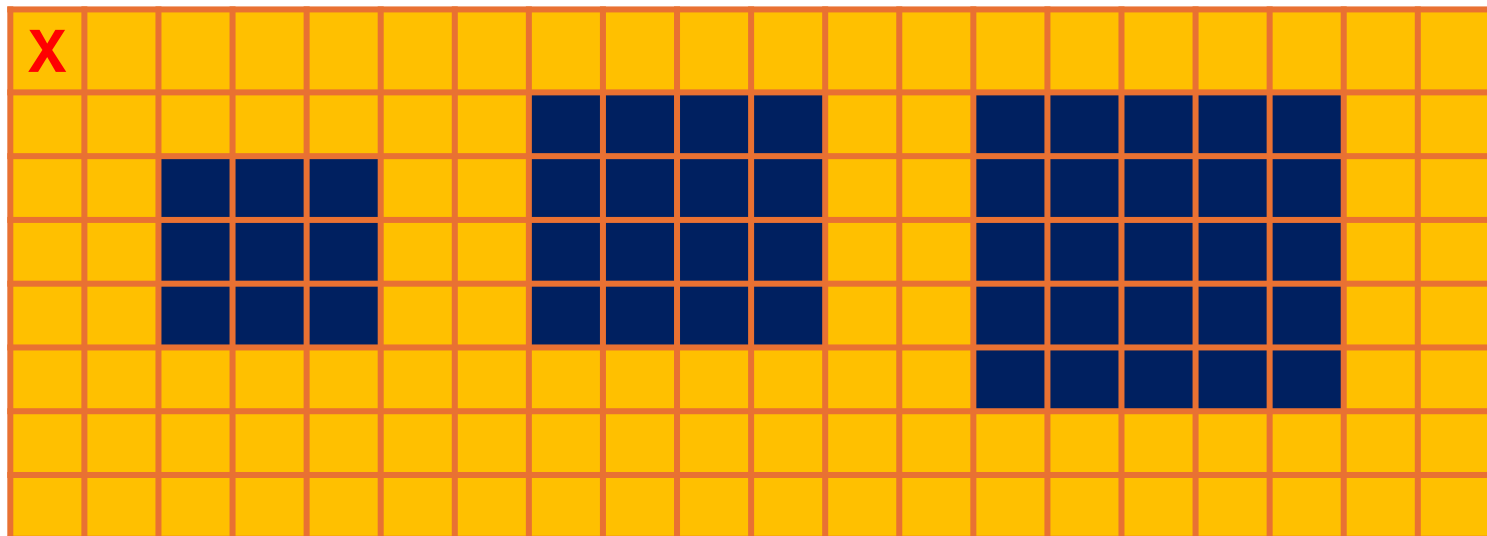
Step 3: Perform the I^c

$$I^c \ominus (W - S)$$

I



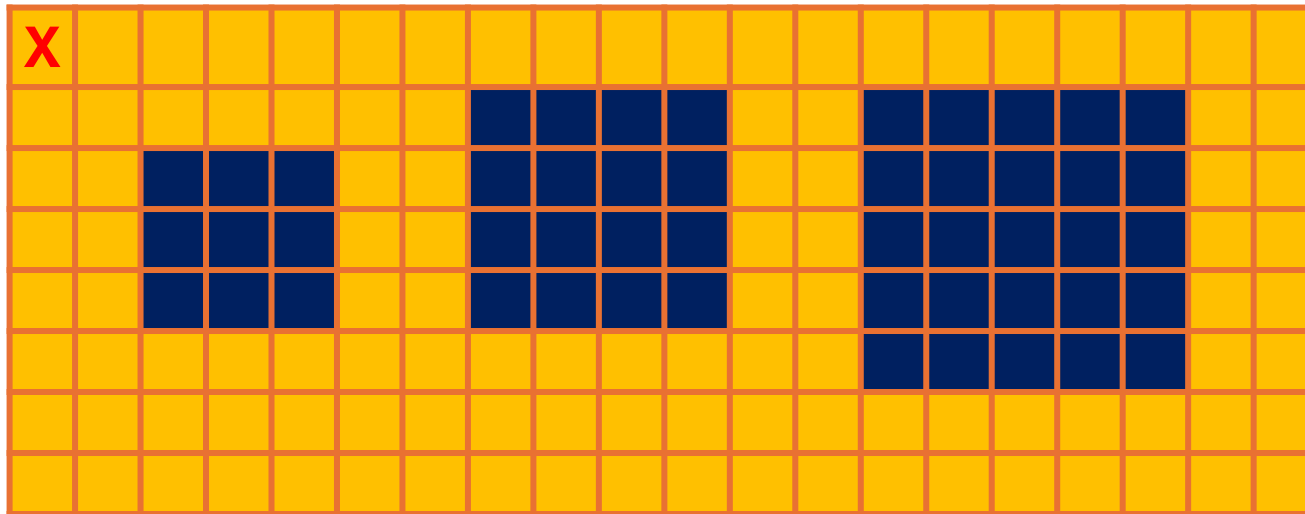
I^c



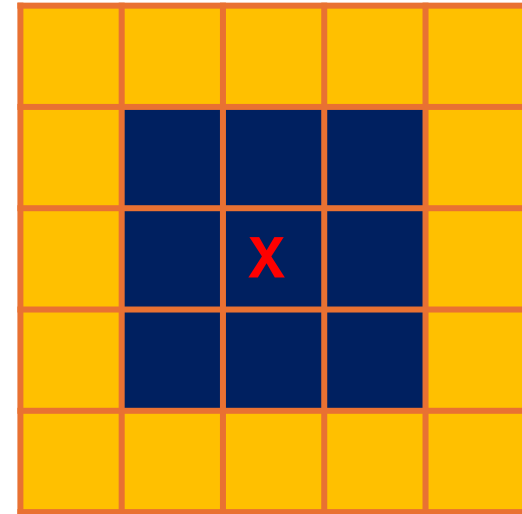
Hit or Miss Transform

Step 4: Determine $(I^C \ominus (W-S))$

I^C

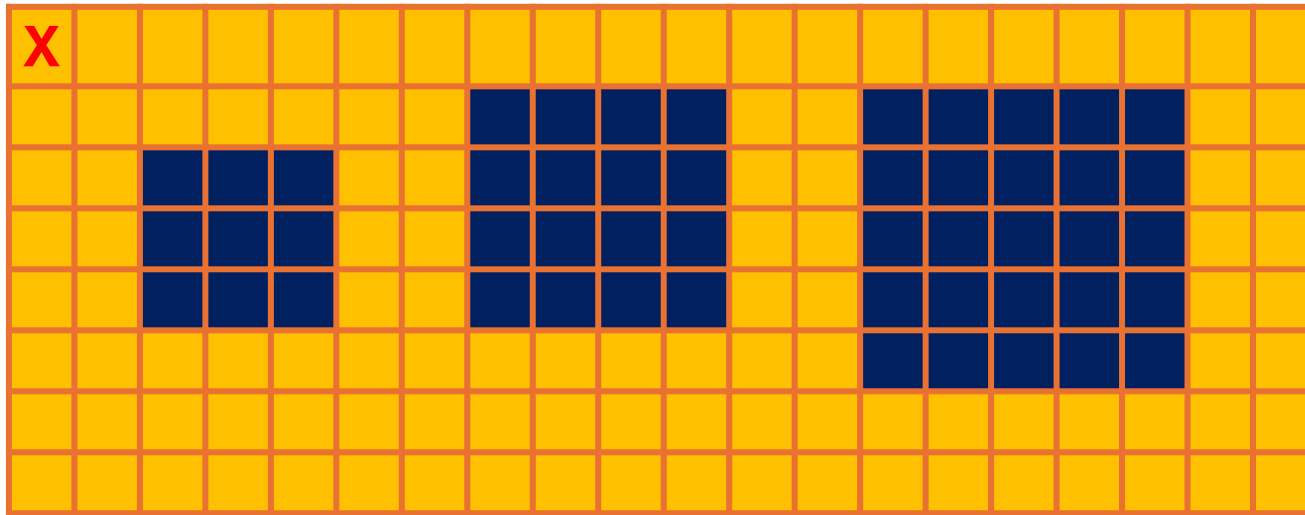


$W-S$

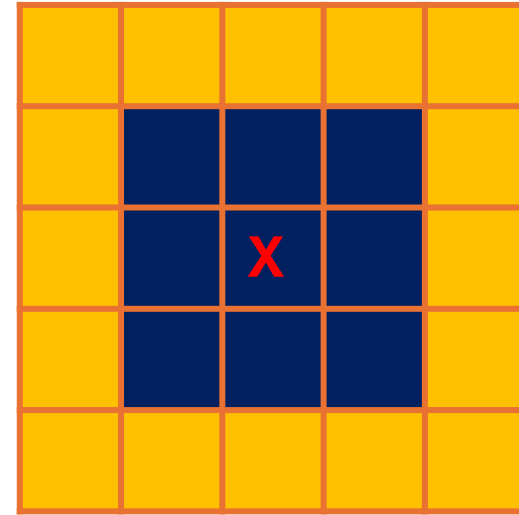


Hit or Miss Transform

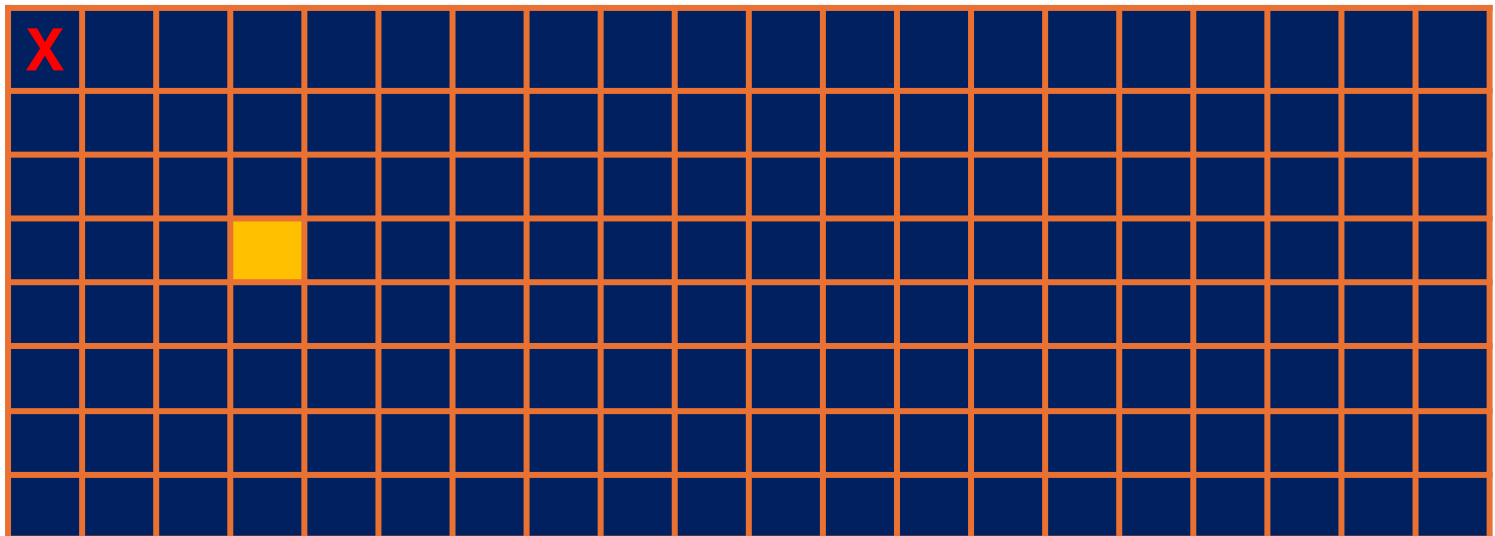
I^c



$W-S$



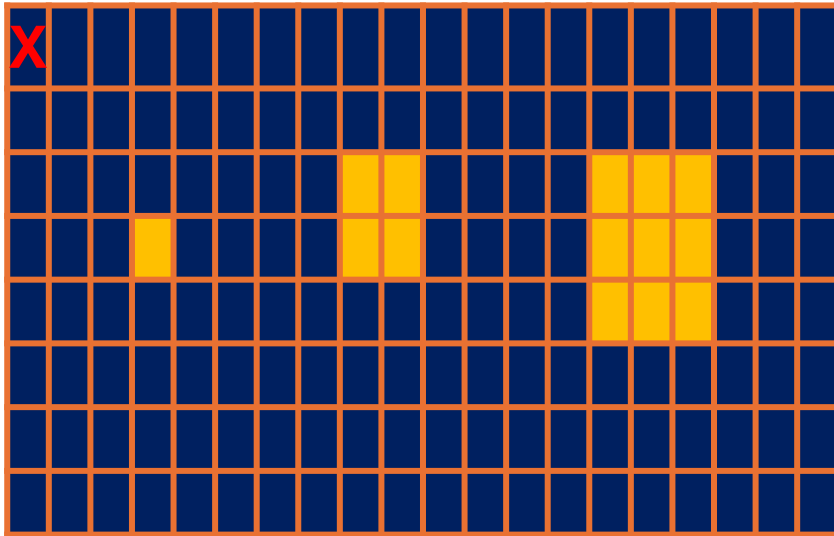
$I^c \ominus (W-S)$



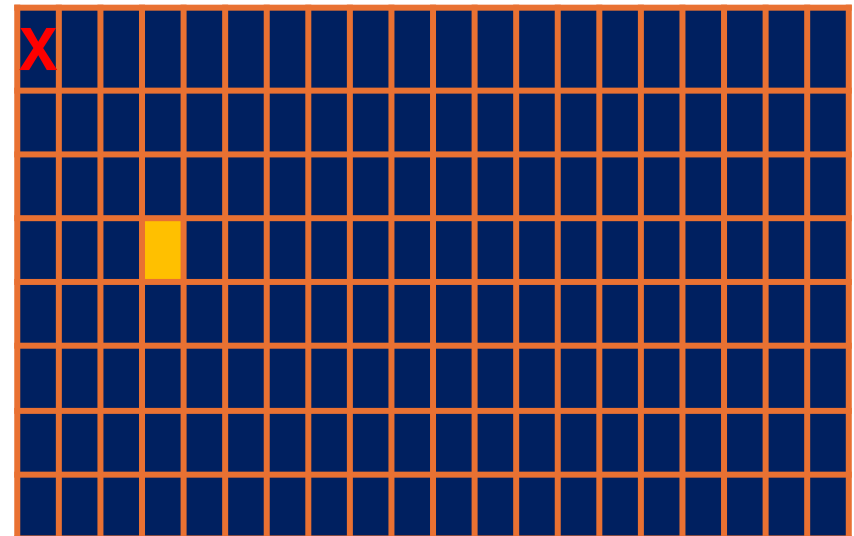
Hit or Miss Transform

Step 4: Perform the operations $(I \ominus S) \cap (I^c \ominus (W - S))$

$$I \ominus S$$

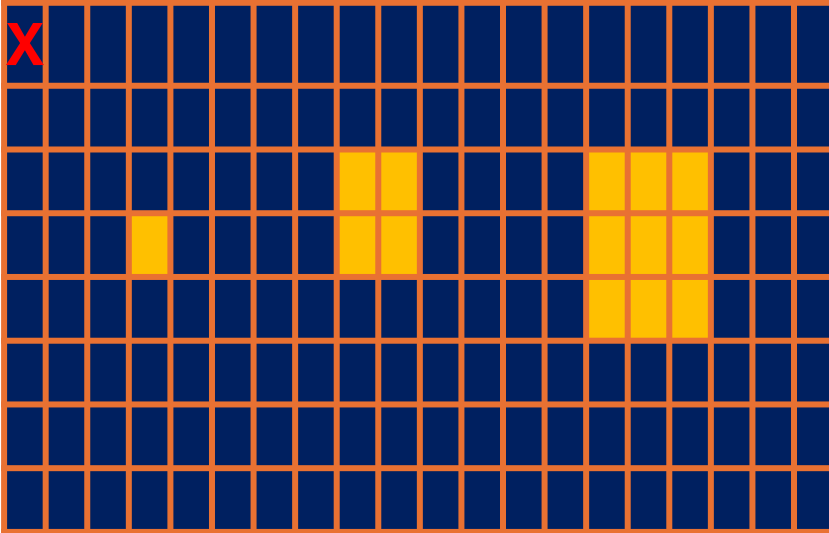


$$I^c \ominus (W - S)$$

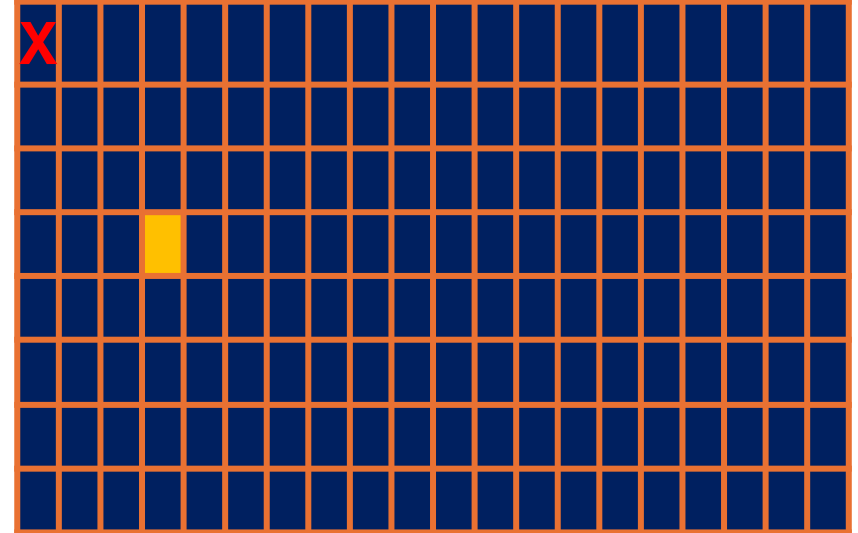


Hit or Miss Transform

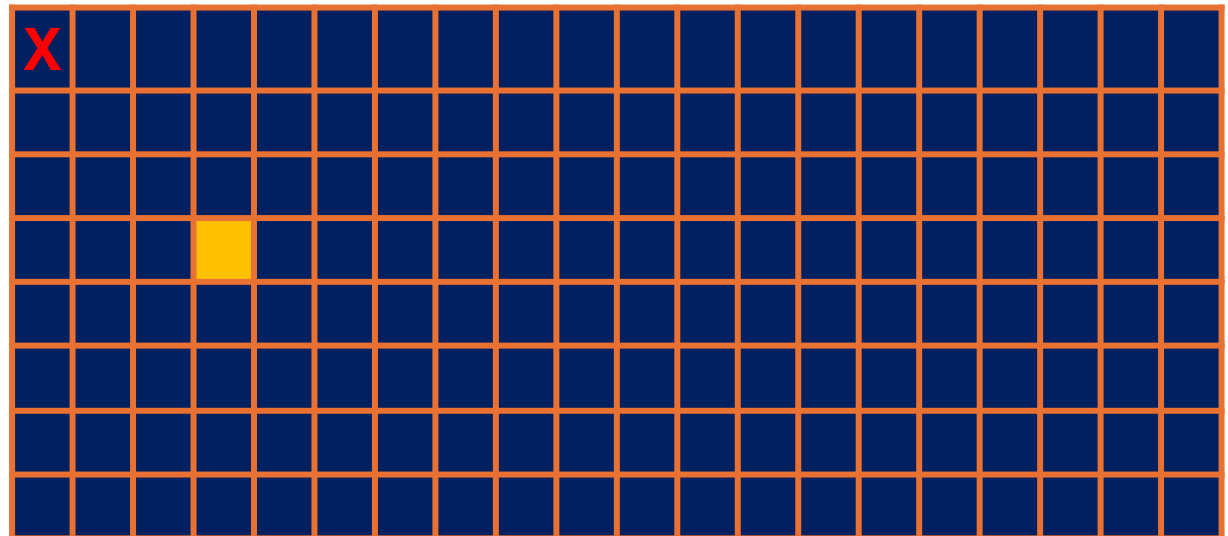
$$I \ominus S$$



$$I^c \ominus (W - S)$$

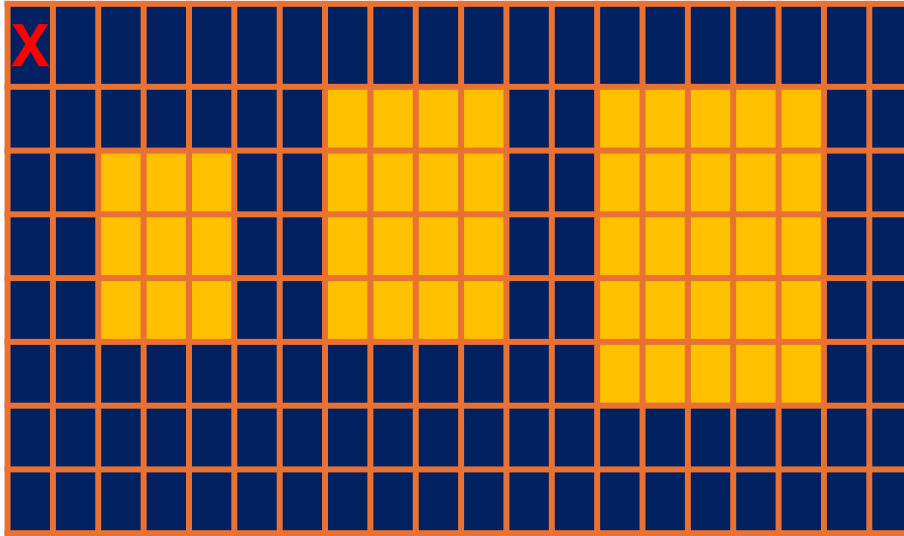


$$(I \ominus S) \cap (I^c \ominus (W - S))$$

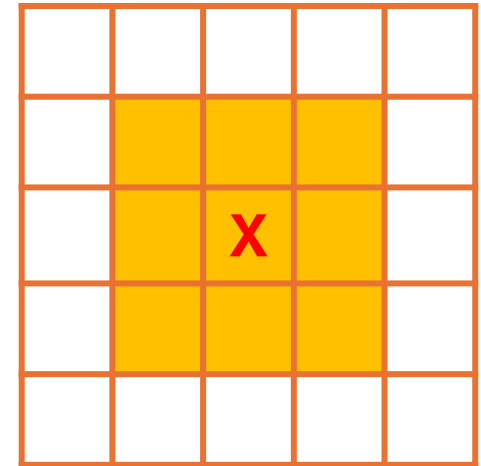


Hit or Miss Transform

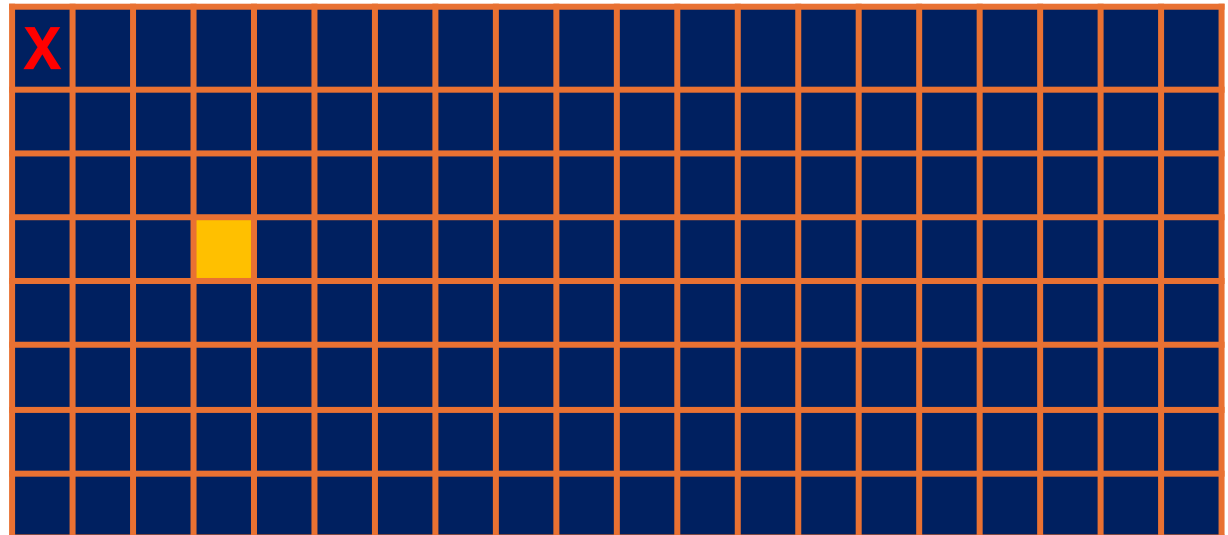
I



S



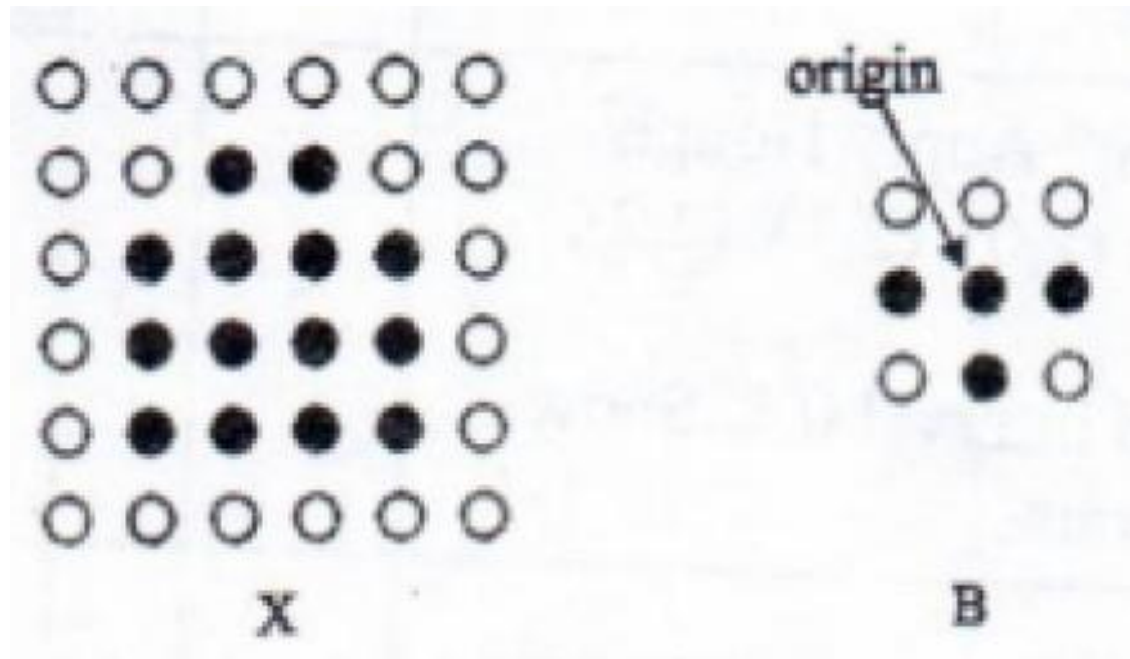
$$(I \ominus S) \cap (I^c \ominus (W - S))$$



University Question

Ques1: Binary Image X and structuring element B, are given in Figure 1. Perform Hit or Miss Transformation. Show all steps.

GLA University Mathura End Term Examination, Even Semester, 2021-22



Boundary Extraction

Boundary Extraction

- ❑ The boundary of an object in an image is the set of pixels that have one or more neighbors
- ❑ A boundary is a ***contour in the image that represents a change in pixel ownership from one object to another***
- ❑ There are two types of boundary
 - **Internal boundary:** It contains boundary pixels that are inside the object

$$\beta(I) = I - (I \ominus S)$$

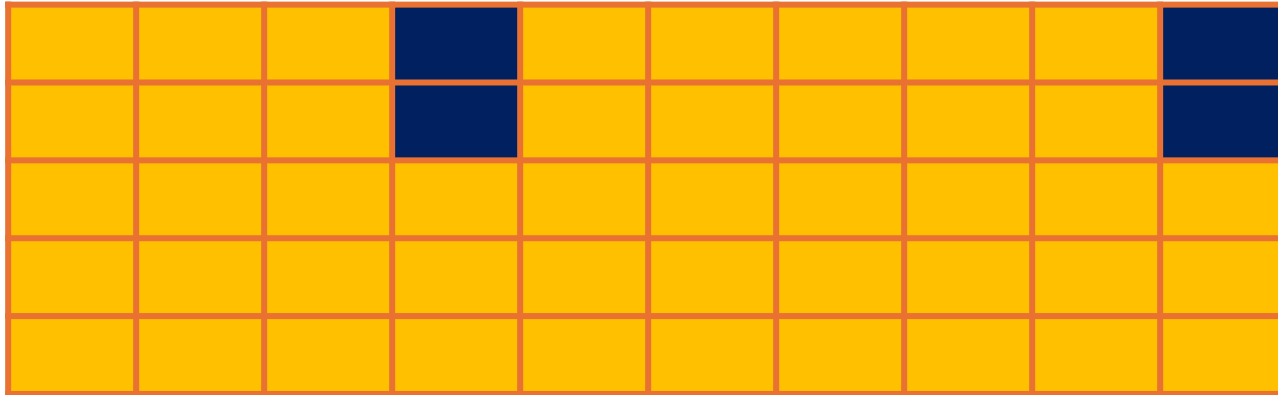
- **External boundary:** It contains boundary pixels that are outside the object

$$\beta(I) = (I \oplus S) - I$$

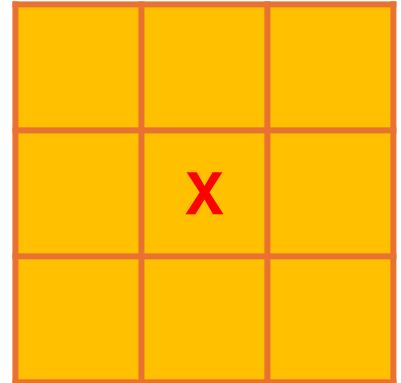
Internal Boundary Extraction

Example 1: Internal Boundary extraction $\beta(I) = I - (I \ominus S)$

I



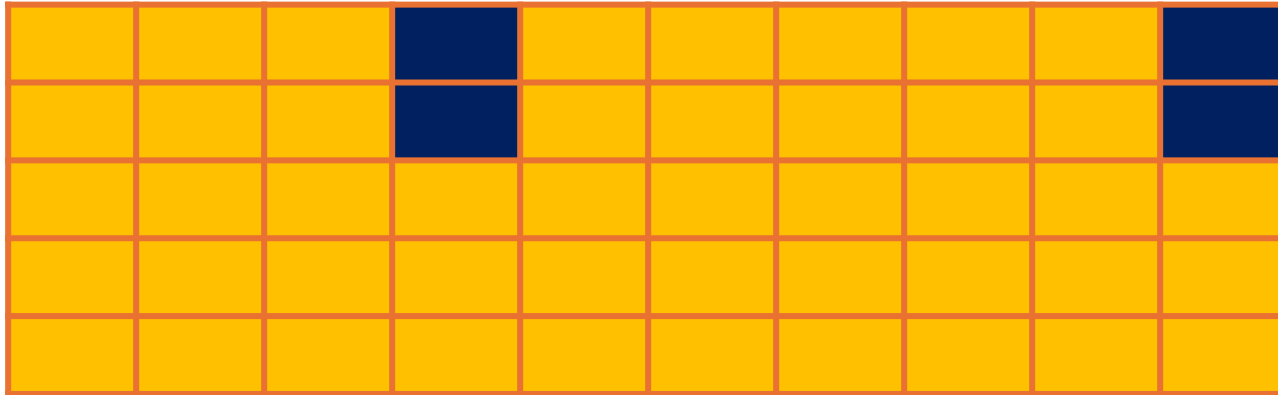
S



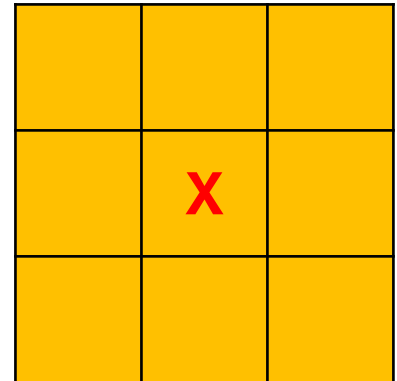
Internal Boundary Extraction

Example 1: Internal Boundary extraction $\beta(I) = I - (I \ominus S)$

I

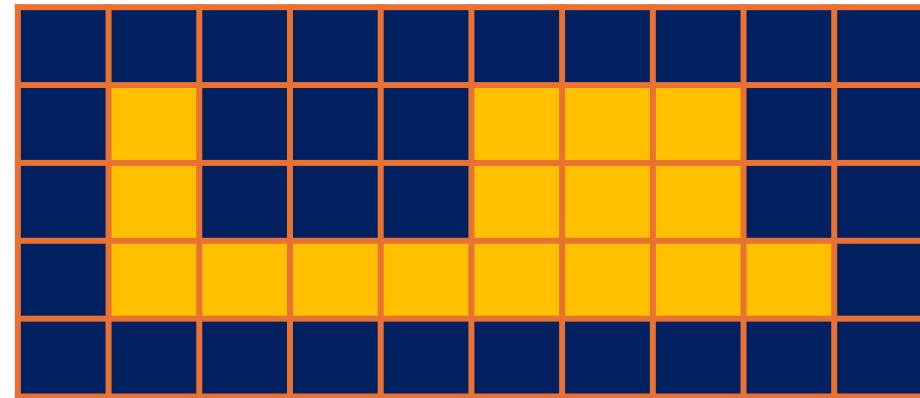
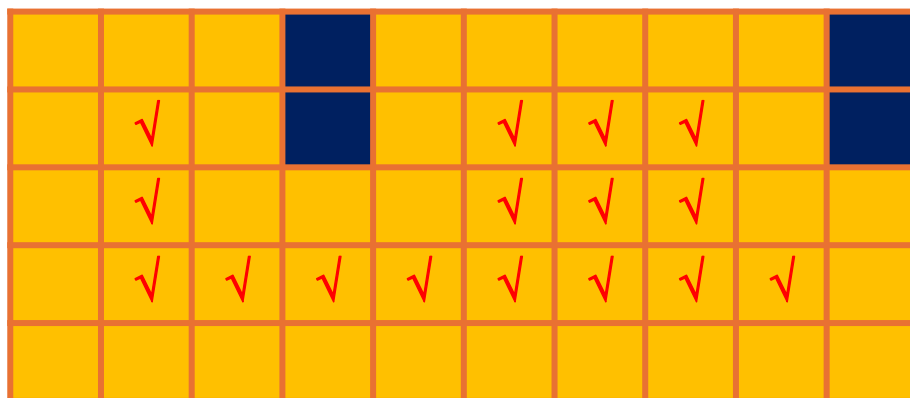


S

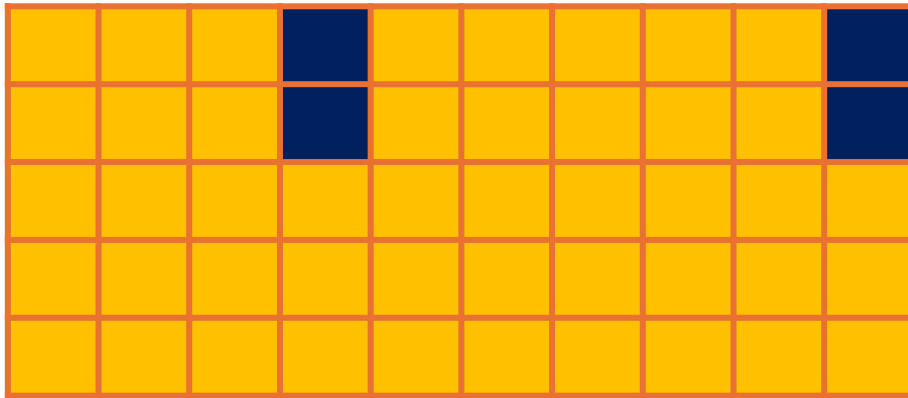
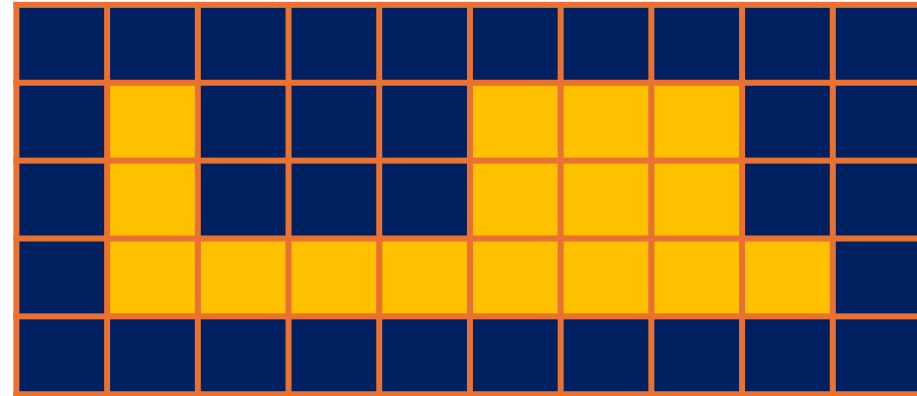


Step 1: Determine $(I \ominus S)$

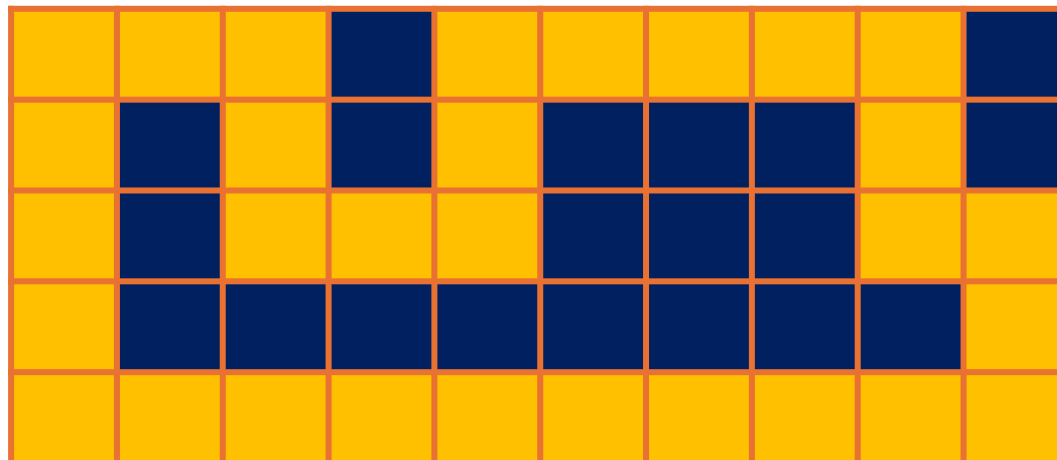
$(I \ominus S)$



I


$$(\mathbf{I} \ominus \mathbf{S})$$


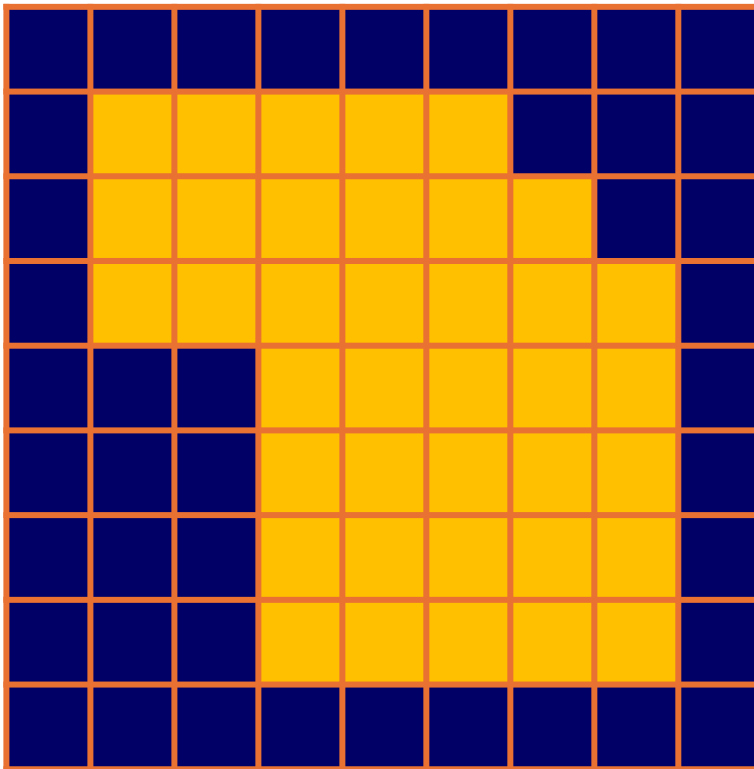
$$\textit{Internal Boundary} = I - (I \ominus S)$$



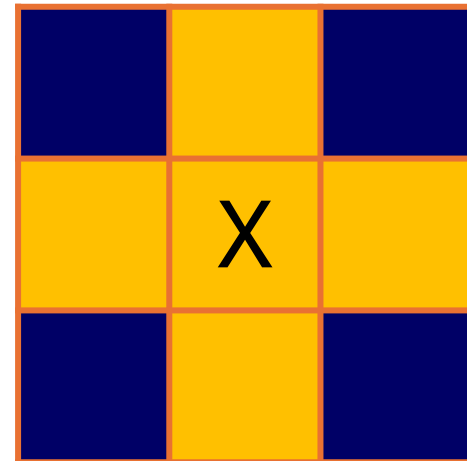
External Boundary Extraction

Example 2: Determine the external boundary $\beta(I) = (I \oplus S) - I$

I



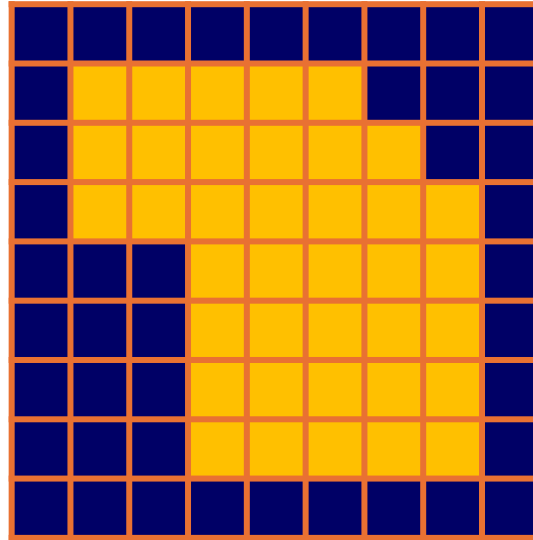
S



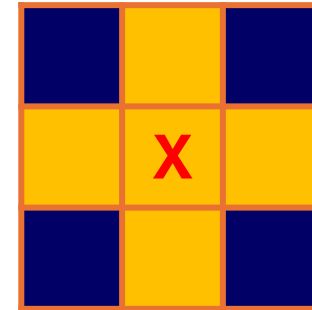
External Boundary Extraction

$$\beta(I) = (I \oplus S) - I$$

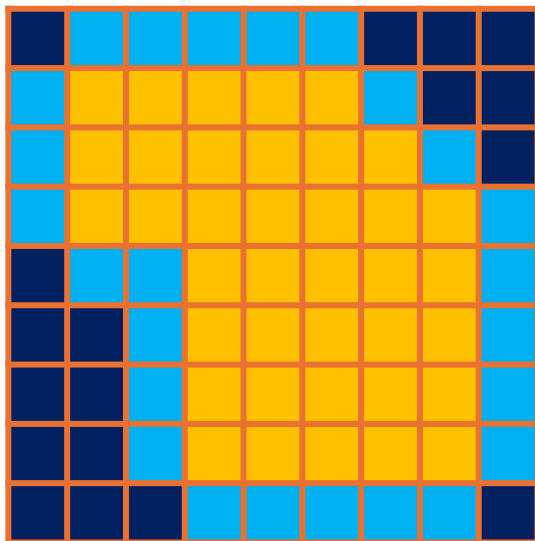
I



S

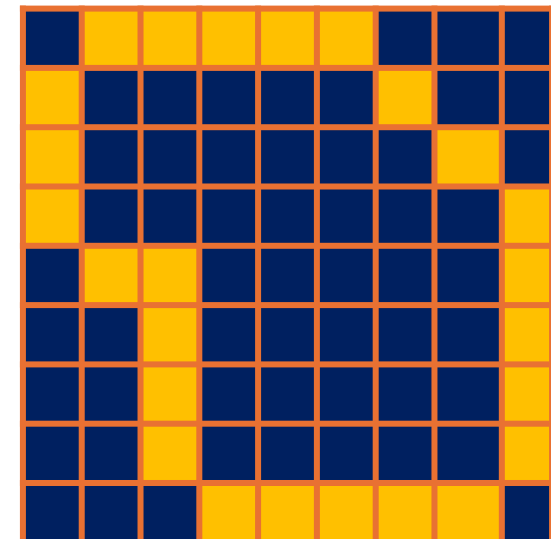
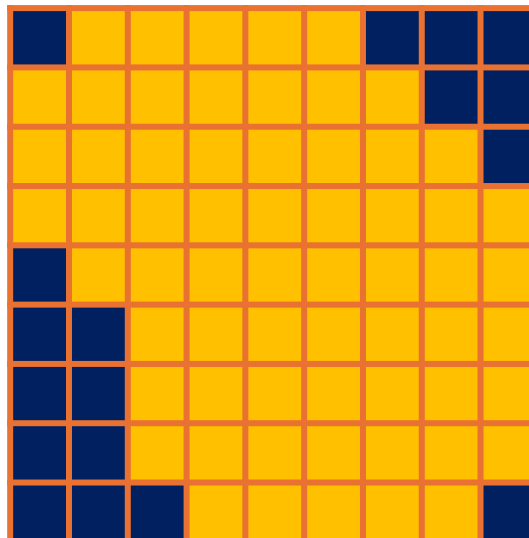


$I \oplus S$



External Boundary

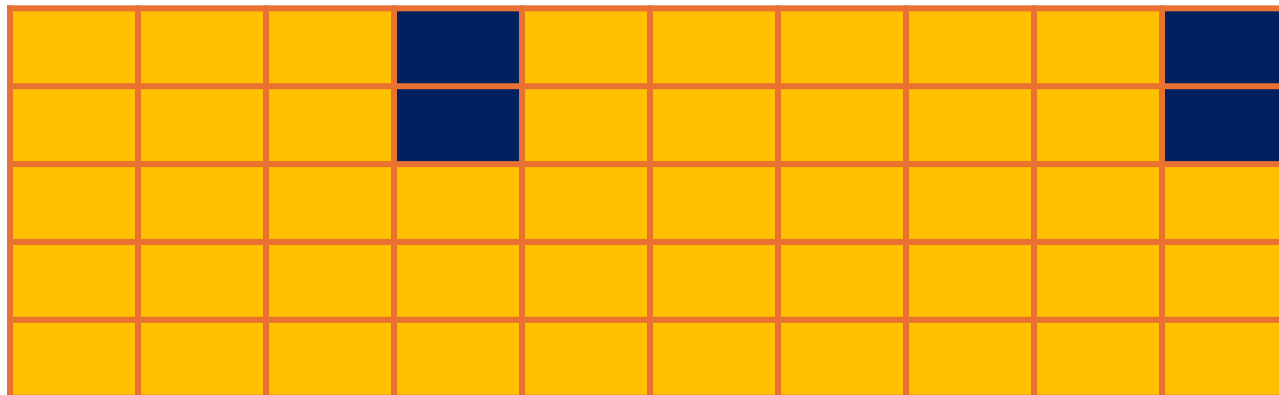
$(I \oplus S) - I$



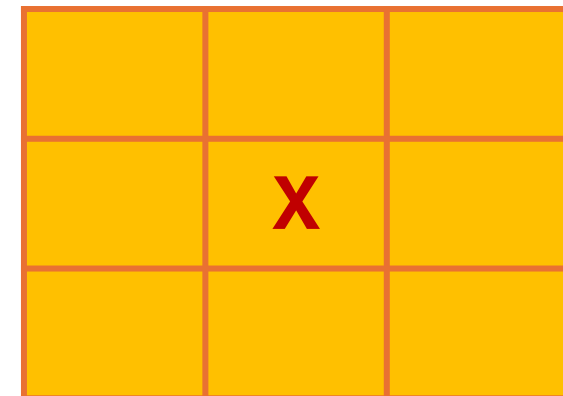
University Question

Ques1: There is an Image A and structuring element B. Write an algorithm over the given image and structuring element to extract the boundary of the image A. (Gray colour represent 1 and White colour represent to 0)

GLA University Mathura End Term Examination, Odd Semester, 2023-24



A



B

University Question

Ques 2: Suppose an image is given in which user is interested to find the pixels that lie inside the boundary? Devise boundary extraction algorithm and find the boundary pixels.

GLA University Mathura End Term Examination, Even Semester, 2023-24

1	1	1	0	1	1	1
1	1	1	0	1	1	1
1	1	1	1	1	1	1
1	1	1	1	1	1	1
1	1	1	1	1	1	1

Image

1	1	1
1	1	1
1	1	1

Structure element

Region Filling

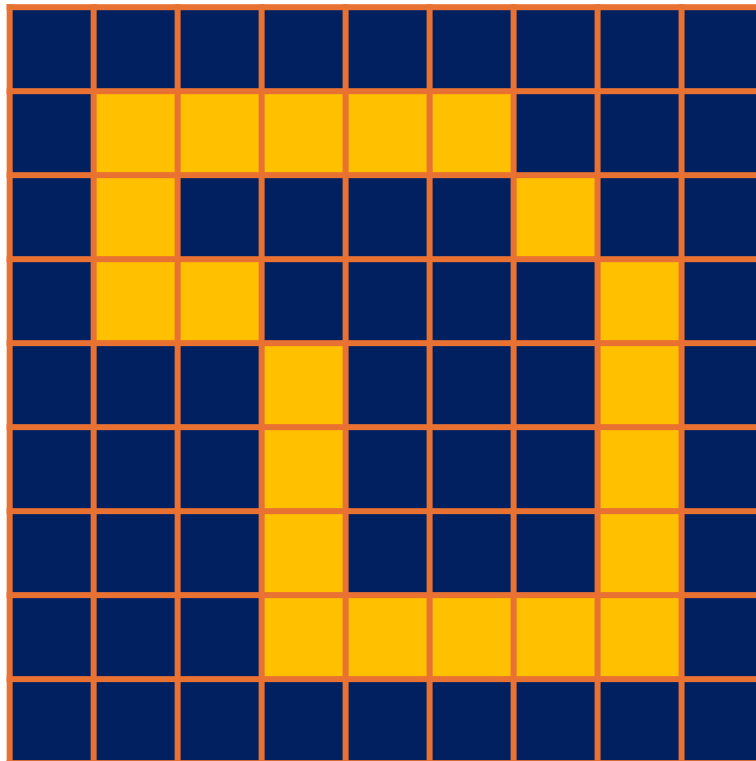
Region Filling

- ❑ Region filling is used to fill the selected region of the object.
- ❑ The process that achieves this is as follows:
 1. Let P be the point of the region to be filled. Initially $P = X_0$. Let A be the subset of element that represents the region. Let K denote the iteration.
 2. Let B be the structuring element.
 3. Repeat steps 4 to 6
 4. Set $K = K + 1$
 5. $X_K = (X_{K-1} \oplus B) \cap A^c$ and store the result
 6. If $X_K = X_{K-1}$ (point of convergence), then STOP
 7. Exit

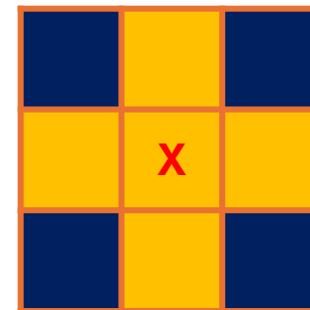
Region Filling

Example 1:

A

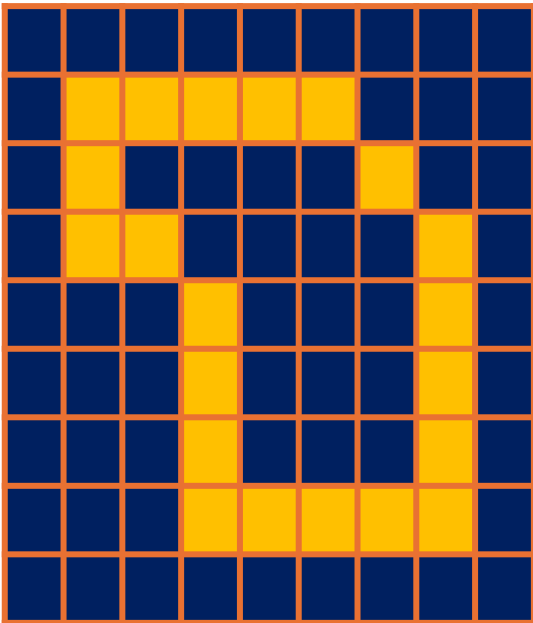


B

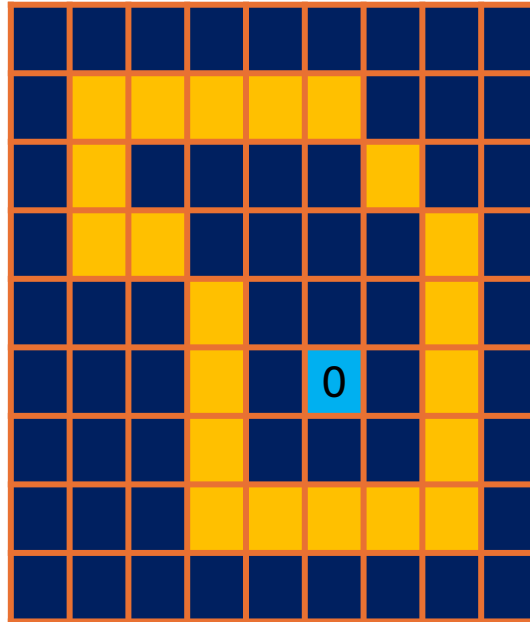


Region Filling

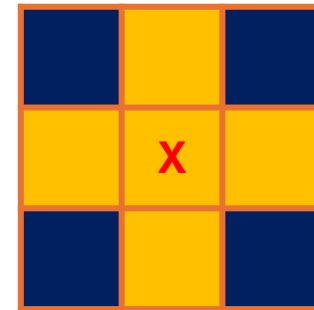
A



X_0



B



Region Filling

$$X_K = (X_{K-1} \oplus B) \cap A^c$$

For K=1

$$X_1 = (X_0 \oplus B) \cap A^c$$

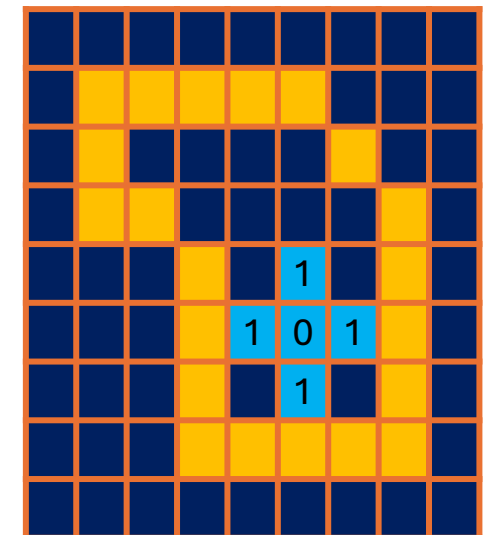
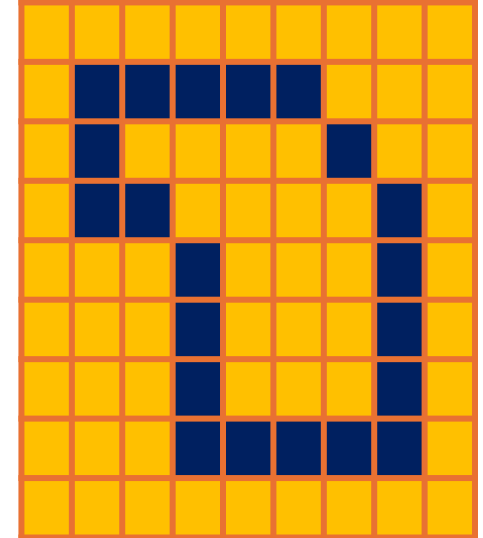
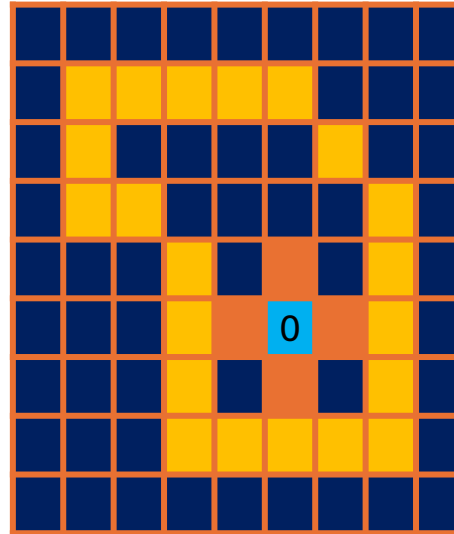
X_0

B

$(X_0 \oplus B)$

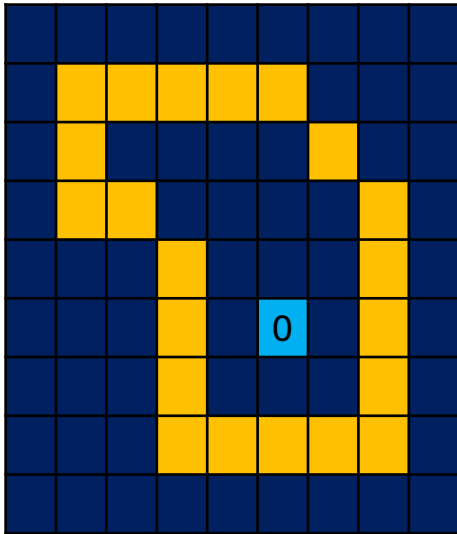
A^c

$$X_1 = (X_0 \oplus B) \cap A^c$$

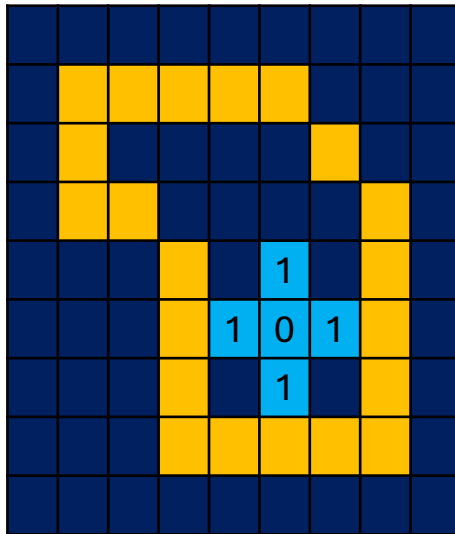


Region Filling

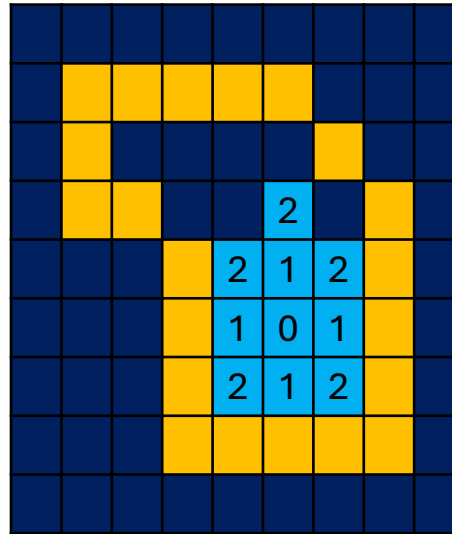
X_0



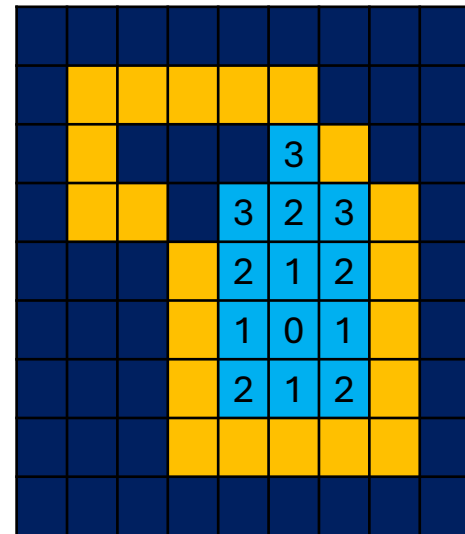
X_1



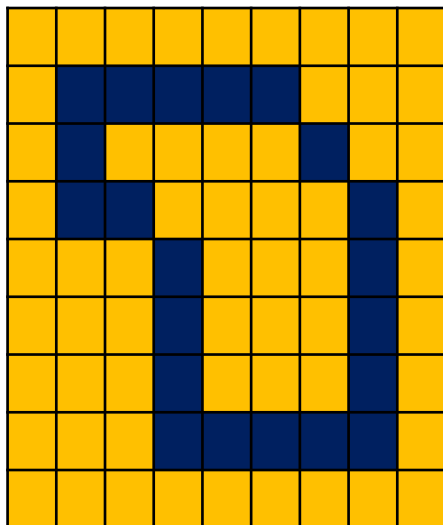
X_2



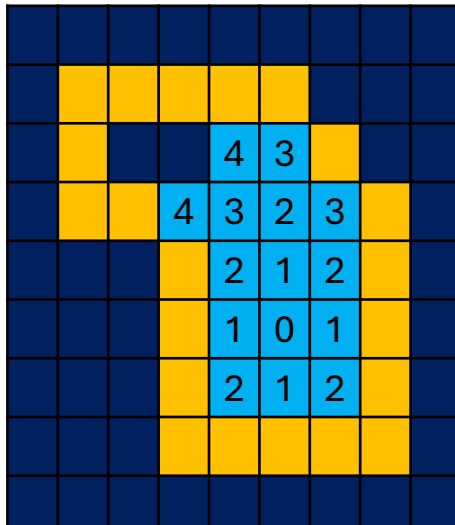
X_3



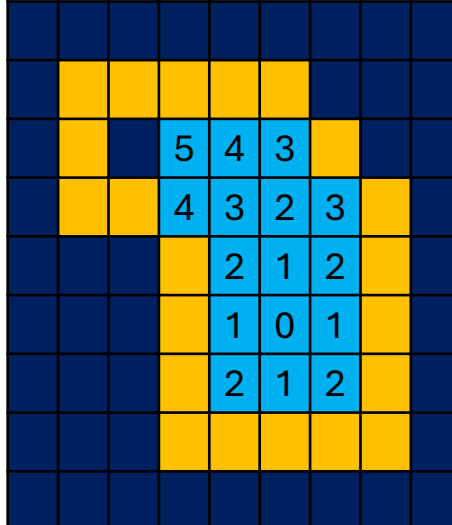
A^c



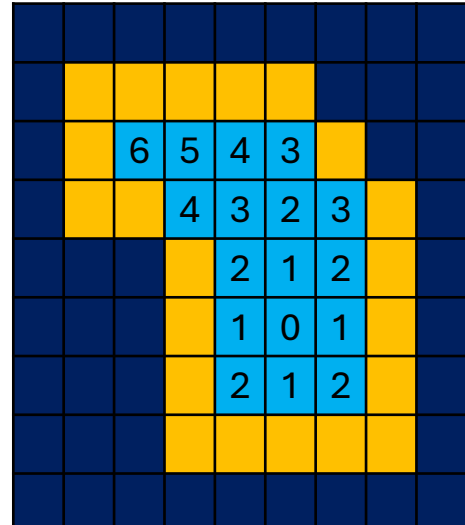
X_4



X_5



X_6



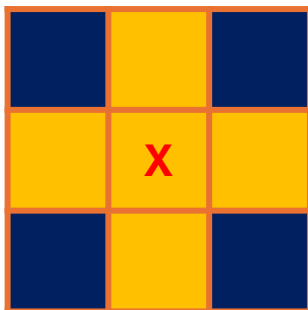
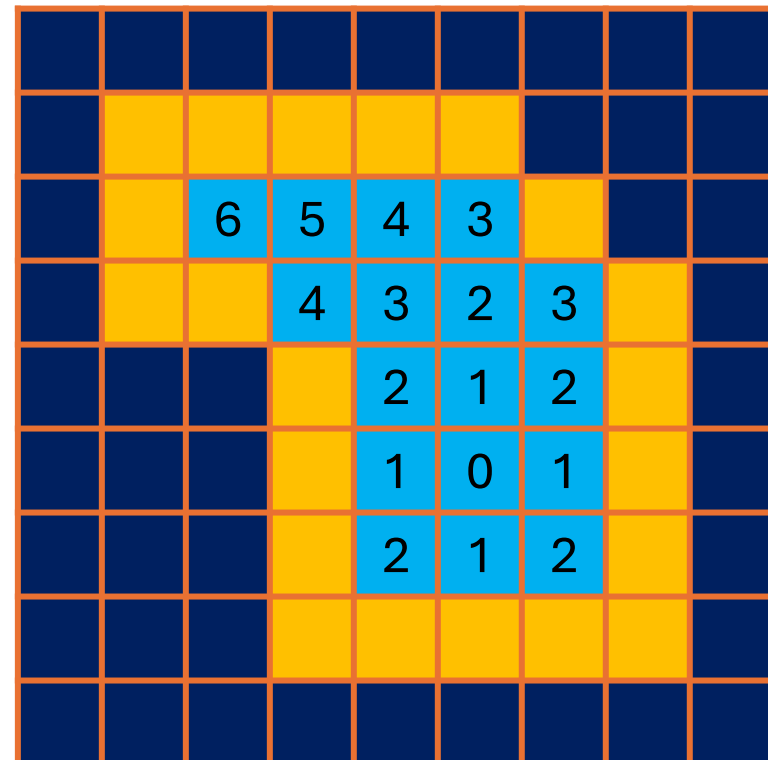
Region Filling

A

A^C

Region Filling Operation

B



University Question

Ques 1: There is an image which consists of regions after segmentation. The regions are having gaps which are to be filled. Apply region filling algorithm to fill region's gap.

GLA University Mathura End Term Examination, Even Semester, 2023-24

		1	1	
	1		1	
	1		1	
	1		1	
	1	1	1	

0	1	0
1	1	1
0	1	0

University Question

Ques 2: There is a segmented Image A in which an object has been detected. The Detected object has boundary pixels, and inner pixels are missing which create a gap. To fill this gap, write an algorithm and apply to fill the inner region of the image A.

GLA University Mathura End Term Examination, Odd Semester, 2023-24

1	1	1	1	
1			1	
1	1		1	
	1		1	
	1	1	1	



Thank You